GG-X-635d
August 13, 1968
SUPERSEDING
Fed. Spec. GG-X-635c
August 21, 1963

FEDERAL SPECIFICATION

# X-RAY APPARATUS SET, RADIOGRAPHIC AND FLUOROSCOPIC, MEDICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification describes the minimum requirements of single phase shockproof radiographic and fluoroscopic apparatus used by the medical profession in diagnostic radiology.

1.2 Classification.

1.2.1 Items, types and classes. The x-ray apparatus covered by this specification shall be of the following items, types, and classes (if applicable), as specified in the invitation for bids.

1.2.1.1 High-tension transformers. Full wave rectified (see 3.2 and 6.1 (b)).

Type I.
Long time rating ...
5 MA at 100 KVP

Momentary rating ...
100 MA at 100 KVP

Class 1 - For energizing one x-ray tube.
Class 2 - For energizing two x-ray tubes.

Type II.
Long time rating ...
5 MA at 125 KVP

Momentary rating ...
200 MA at 125 KVP

Class 1 - For energizing one x-ray tube.
Class 2 - For energizing two x-ray tubes.

Type III.
Long time rating ...
5 MA at 125 KVP

Momentary rating ...
300 MA at 125 KVP

Class 1 - For energizing one x-ray tube.
Class 2 - For energizing two x-ray tubes.
Class 3 - For energizing three x-ray tubes.

Type IV.
Long time rating ...
5 MA at 130 KVP

Momentary rating ...
300 MA at 150 KVP

Class 1 - For energizing one x-ray tube.
Class 2 - For energizing two x-ray tubes.
Class 3 - For energizing three x-ray tubes.



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Type V.

Long time rating ...

5 MA at 125 KVP

Momentary rating ...

500 MA at 125 KVP and
300 MA at 150 KVP

Class 1 - For energizing one x-ray tube.
Class 2 - For energizing two x-ray tubes.
Class 3 - For energizing three x-ray tubes.

Type VI.

Long time rating ...

5 MA at 130 KVP

Momentary rating ...

500 MA at 150 KVP

Class 1 - For energizing one x-ray tube.
Class 2 - For energizing two x-ray tubes.
Class 3 - For energizing three x-ray tubes.

Type VII.

Long time rating ...

5 MA at 125 KVP

Momentary rating ...

800 MA at 110 KVP and
500 MA at 150 KVP

Class 1 - For energizing one x-ray tube.
Class 2 - For energizing two x-ray tubes.
Class 3 - For energizing three x-ray tubes.

Definitions:

Long time rating: A long time rating is the rating based on an operating interval of five minutes or longer.

Momentary rating: A momentary rating is the rating based on an operating interval that does not exceed 5 seconds.

1.2.1.2 Control cabinets (see 3.3 and 6.1 (c)).

Type I - Console.
Type II - Vertical-panel cabinet.
Type III - Wall mounted.

1.2.1.3 Tube stands (see 3.4 and 6.1 (d)).

Type I - Tube stand supported from overhead only.
Type II - Tube stand supported from floor and overhead.

1.2.1.4 Tables (see 3.5 and 6.1 (e)).

Type I - Nontilting.
Class 1 - Horizontal bucky table.
Class 2 - Pedestal base table.

Type II - Enclosed tilt table with 15° Trendelenburg.
Class 1 - Radiographic only 15° tilt table with bucky (no fluoroscopic facilities).
Class 2 - Fluoroscopic only 15° tilt table with automatic movement spotfilm tunnel (no bucky diaphragm).

Style (a) Stationary tabletop.
Style (b) Tabletop with a two-way power driven longitudinal travel.



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Class 3 - Radiographic-fluoroscopic 15° tilt table with semiautomatic movement spotfilm tunnel.

Style (a) Stationary tabletop.
Style (b) Tabletop with a two-way power driven longitudinal travel.

Class 4 - Radiographic-fluoroscopic 15° tilt table with automatic movement spotfilm tunnel for use with 8 by 10 inch cassette.

Style (a) Stationary tabletop.
Style (b) Tabletop with a two-way power driven longitudinal travel.

Class 5 - Radiographic-fluoroscopic 15° tilt table with automatic movement spotfilm tunnel for use with 9-1/2 by 9-1/2 inch cassette with glide back or horizontal parking and automatic programmed shutters.

Style (a) Stationary tabletop.
Style (b) Tabletop with a two-way power driven longitudinal travel.

Class 6 - Radiographic-fluoroscopic tilt table with image intensifier support, circular collimation, glide back or horizontal parking, without spotfilm tunnel.

Style (a) Stationary tabletop.
Style (b) Tabletop with two-way power driven longitudinal travel.

Type III - Enclosed tilt table with more than 15° Trendelenburg. (Degree of Trendelenburg will be indicated in the invitation for bid.)

Class 1 - Radiographic-fluoroscopic tilt table with automatic movement spotfilm tunnel for use with 8 by 10 inch cassettes with hinged or vertical parking.

Style (a) Stationary tabletop.
Style (b) Tabletop with a two-way power driven longitudinal travel.
Style (c) Tabletop with a four-way movement, power driven longitudinal and either manual or power lateral travel.

Class 2 - Radiographic-fluoroscopic tilt table with automatic movement spotfilm tunnel for use with 9-1/2 by 9-1/2 inch cassettes, with glide back or horizontal parking and automatic programmed shutters.

Style (a) Stationary tabletop.
Style (b) Tabletop with a two-way power driven longitudinal travel.
Style (c) Tabletop with a four-way movement, power driven longitudinal and either manual or power lateral travel.

Class 3 - Radiographic-fluoroscopic tilt table with image intensifier support, circular collimation, glide back or horizontal parking, without spotfilm tunnel.

Style (a) Stationary tabletop.
Style (b) Tabletop with two-way power driven longitudinal travel.
Style (c) Tabletop with a four-way movement, power driven longitudinal and either manual or power lateral travel.

1.2.1.5 Body sectional devices (see 3.6 and 6.1 (f)).

Type I - For horizontal body sectional radiography.
Type II - For vertical body sectional radiography.
Type III - For a combination of both horizontal and vertical body sectional radiography.

1.2.1.6 X-Ray tubes (see 3.7 and 6.1 (g)).

Type I - Rotating-anode radiographic tube.
Class 1 - 125 KVP rating.
Class 2 - 150 KVP rating.

Style (a) Conventional insert.
Style (b) High speed insert.



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Type II - Rotating-anode fluoroscopic tube.

Class 1 - 125 KVP rating.
Class 2 - 150 KVP rating.

Style (a) Conventional insert.
Style (b) High speed insert.

1.2.1.7 Operating-room apparatus (see 3.1.3).

1.2.1.8 Cables. For high-tension transformers, control, and x-ray tubes (see 3.8):

(a) Shockproof.
(b) X-Ray tube blower.
(c) X-Ray tube rotor.
(d) Stereoscopic shift.
(e) Centering light.
(f) Other cables as may be required in the invitation for bids.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055- Barrier Material, Waterproofed, Flexible.

Federal Standards:

Fed. Std. No. 72 - Shockproof Cable Terminal and Receptacle for Use On X-Ray Equipment.
Fed. Std. No. 76 - Cassettes, Radiographic Film (Medical), Dimensions.
Fed. Std. No. 83 - X-Ray Tube Focal Spot, Method of Measurement.
Fed. Std. No. 123- Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-C-104 - Crates Wood, Lumber and Plywood Sheathed, Nailed and Bolted.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

United States of America Standards Institute:

B1.1-1960 - Unified Screws Threads.



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(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

Underwriters' Laboratories (UL) Inc., Standards:

No. 187 - Standards for Safety, X-Ray Equipment.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, Long Island, New York, 11746; 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California, 95050.)

National Fire Protection Association (NFPA) Publications:

No. 56 - Code for the Use of Flammable Anesthetics.
No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston Mass., 02110.)

National Council on Radiation Protection and Measurements:

NCRP Report No. 33

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Materials. Satisfactory high-grade materials shall be used throughout. All parts entering into the construction of the x-ray equipment covered by this specification shall be fabricated from materials which are entirely satisfactory for the purpose and of sufficient strength to maintain operational alignment for the life of the equipment. The equipment shall be free from defects which may be injurious, detract from its appearance, or that may impair its serviceability.

3.1.1 Fire and casualty hazards.

3.1.1.1 Each bidder shall submit to the contracting agency proof that the equipment he proposes to supply under this specification conforms to the applicable requirements of UL No. 187. The label, or listing with re-examination of the Underwriters' Laboratories, Inc., may be accepted as evidence that the equipment conforms to this requirement.

3.1.1.1.1 In lieu of the label or listing with re-examination, the bidder may submit independent proof satisfactory to the contracting agency that the equipment conforms to the applicable requirements of the published standards, including methods of tests, of UL No. 187.

3.1.1.1.2 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.1.1.2 Radiation protection. The apparatus offered shall be in accordance with the requirements of NCRP Report No. 33. Any applicable sentence regarding x-ray apparatus using the words "shall", "should", or "advisable" shall be considered a requirement.

3.1.2 Construction. All exposed welded parts shall be ground free of scale and protuberances to provide the best possible surface for paint or plating. All parts shall be suitably protected against corrosion by painting, plating, or equivalent means. The general finish of the equipment shall be enamel in the manufacturer's standard color. All exposed parts subject to wear and handling shall have a hard durable wearing finish. All bolt connections shall be secured by lockwashers. Plastic parts such as knobs shall have molded metal reinforcements if set screws are utilized. Whenever applicable, screw threads shall be of the Unified Thread Series of USASI Standard B1.1-1960.

3.1.3 Operating room equipment. Operating rooms are subjected to explosive atmospheres when gases are used as anesthetic agents. The x-ray tube and shockproof cables shall be listed by the Underwriters' Laboratories, Inc. as being safe to use in class 1, group C atmospheres, when used with specified x-ray controls and high-tension transformers. The control cabinets, high-tension transformers and tubestands shall be positioned outside of the hazardous area and the installation shall comply with the requirements of NFPA No. 56, and Article 517 of NFPA No. 70.

3.1.4 Generator performance. Each generator, the combination of the control and transformer specified, when tested for performance as described in 4.4, shall meet the performance requirements listed in 4.4. Reproducibility and radiographic consistency shall meet the requirements of



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3.3.3.2.4.2, where applicable, and the requirements listed in 4.4.3.12, when determined as described in 4.4.3.12.

3.1.4.1 Adaptability. Each generator shall be capable of being readily adaptable for use with an image intensifier and types III, IV, V, VI, and VII generators with cine radiographic equipment and television chains.

3.2 High-tension transformer.

3.2.1 Capacity. The high-tension transformer, when used in conjunction with its control cabinet, shall be capable of delivering in accordance with the ratings specified for the individual types and classes of transformers as referenced in paragraph 1.2.1.1.

3.2.2 High-tension transformer assembly. All parts of the high-tension transformer assembly necessary for converting the applied primary alternating-current into full wave, rectified high-tension voltage, and the furnishing of the proper cathode energy to the appropriate pair of shockproof cable receptacles for the class of high-tension transformer enumerated in the invitation for bids shall be contained in an appropriate metal container and immersed in a high-grade insulating medium. The assembly shall provide for concealing the terminal connections. All electrical conductors used inside, on, or connecting to the high-tension assembly shall employ insulation which will not deteriorate as a result of contact with the insulating medium.

3.2.2.1 High potential requirement. The high-tension transformer shall be capable of withstanding without breakdown for a period of one minute, the application to the primary winding, an alternating potential of rated frequency and equal to the no-load voltage required to produce 125 percent of the long time secondary voltage rating. The test shall be conducted as described in 4.2.

3.2.2.2 High-tension rectifiers. The high-tension rectifier or rectifiers shall convert high-tension alternating-current voltage to high-tension full wave rectified voltage, and shall not interfere with radio communications. The rectifiers may be valve tubes or high-voltage semiconductors, as specified in the invitation for bid. The rectifiers shall be warranted to be free of manufacturing defects which impair their operating life for a period of 10,000 hours (filament life) or 5 years, whichever comes first. A means shall be provided to permit single impulse exposures for radiography and cine radiography when types III, IV, V, VI, and VII transformers are specified. Means shall be provided to prevent transformer saturation from unidirectional current flow (see 6.1 (z)).

3.2.2.3 Ease of replacement of high-voltage rectifier. The high-tension transformer assembly shall be provided with a means for easy replacement, servicing, and inspection of the high-voltage rectifier or rectifiers. If transparent handhole covers and "bull's eyes" are provided for replacement or inspection of the valve tubes, additional covers shall be provided to prevent illumination from the valve tube filaments from leaving the high-tension transformer assembly. It shall not be necessary "to pull" the transformer to replace the rectifiers.

3.2.2.4 High-tension switch. The high-tension switch shall be furnished in type I, class 2, and types II, III, IV, V, VI, and VII, classes 2 and 3 high-tension transformers. This switch shall be electrically operated and remotely controlled at the control cabinet and shall make it possible to deliver the proper cathode energy and full-wave, rectified, high-tension to the x-ray tube through shockproof cables connected to any selected pair of shockproof cable receptacles.

3.2.2.5 Shockproof cable receptacles. Each high-tension transformer assembly shall be supplied with one pair of shockproof cable receptacles for each x-ray tube that the high-tension transformer is to energize. These shockproof cable receptacles shall be in accordance with Fed. Std. No. 72, and the high-tension transformer assembly shall be permanently marked "Equipped with Federal Standard Shockproof Cable Receptacles". Covers shall be supplied for receptacles not in use.

3.2.2.5.1 Cathode cable receptacle. One shockproof cable receptacle of each pair shall provide proper cathode energy to either filament of a double focus x-ray tube. This receptacle shall be permanently labeled "Cathode".

3.2.2.5.2 Anode cable receptacle. The other shockproof cable receptacle of each pair shall provide for electrical connection of the anode of the x-ray tube to high-tension transformer assembly by means of a shockproof cable. This receptacle shall be permanently labeled "Anode".

3.3 Control cabinets. Control cabinets shall be of the following types:

(a) Console.
(b) Vertical-panel.
(c) Wall-mounted.



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3.3.1 Definitions:

3.3.1.1 Console control cabinet. A console control cabinet is defined as one occupying a floor area not greater than 4 sq. ft. with a width not exceeding 36 inches and whose height is not less than 3 ft. nor greater than 5 ft., using the flat or slanted top for placement of the operational controls.

3.3.1.2 Vertical-panel control cabinet. A vertical-panel control cabinet is defined as one whose floor dimensions do not exceed 4-1/2 sq. ft. with a width not exceeding 36 inches and whose height is not less than 4 ft. 6 inches, nor greater than 6 ft. 6 inches, with front mounted controls.

3.3.1.3 Wall-mounted control cabinet. A wall-mounted control cabinet is one housing identical components as the console or vertical-panel control cabinet and mounted either flush or on the wall and not exceeding 36 inches in width.

3.3.2 Other type control cabinets. Should a control cabinet whose shape and style is other than the types described herein be desired, detailed information regarding same shall be specified in the invitation for bids.

3.3.3 Common control cabinet requirements. There are several types of generators covered by this specification. The requirements for control cabinets to function with each type generator vary. The requirements common to all control cabinets are specified in the following paragraphs.

3.3.3.1 Contents of control cabinet. The control cabinet shall contain all parts for controlling the various components of equipment in this specification, plus any other parts necessary for the coordinated action of the various components and electrical circuits as described therein. Components may have adjustments and regulators incorporated within the component proper, however, the electrical energy shall be obtained from within the control cabinet proper. Specifically, each control shall contain the following items:

(a) Mail-line switch (see 3.3.3.1.1).
(b) Circuit breaker (see 3.3.3.1.2).
(c) Input voltage circuit (see 3.3.3.1.3).
(d) Line compensator device (see 3.3.3.1.4).
(e) Milliampere technique selector switch (see 3.3.3.1.5).
(f) Kilovoltage control or controls (see 3.3.3.1.6).
(g) Prereading kilovolt indicator (see 3.3.3.1.7).
(h) Milliammeter (see 3.3.3.1.8).
(i) Radiographic timer (see 3.3.3.1.9).
(j) Tube indicator when required by 3.3.3.1.10.
(k) Rotor control apparatus, (except high-speed frequency converter and adaption circuitry) (see 3.3.3.1.11 and 3.3.3.2.6).
(l) Foot switch circuit (see 3.3.3.1.12).
(m) MA stabilizer (see 3.3.3.1.13).
(n) Bucky switching (see 3.3.3.1.14).
(o) X-ray tube protective circuits in controls for use with types III, IV, V, VI, and VII transformers (see 3.3.3.2.6).
(p) Rotor speed changing equipment for use with types IV, V, VI, and VII high-tension transformers (see 3.3.3.1.11 (c)).

3.3.3.1.1 Main-line switch shall break both sides of the line energizing the equipment. The action of this switch shall provide for quick-make and break-of the circuit with a minimum of arcing and burning of contact points. The design shall be to eliminate the possiblity of "partial contact".

3.3.3.1.2 Circuit breaker. A circuit breaker, effective regardless of the position of the milliamperage technique selector switch, which meets the applicable requirements of the UL No. 187, shall be supplied. The circuit breaker shall be capable of being reset and may be supplied as a component part of the main-line switch. Other branch circuits shall be properly protected with cartridge-type fuses and arranged so that no exposure can be made if any of the appropriate exposure control circuits are not energized (see 4.3).

3.3.3.1.3 Input voltage circuit. The x-ray equipment provided shall be capable of proper operation when connected to an alternating current source of voltage ranging from 190 to 250 volts for types I, II, III, IV, V, and VI transformers, or 380 to 500 volts for type VII transformers. This power source shall not have a daily line voltage variation, due to causes other than the x-ray apparatus load, exceeding $\pm$ 2-1/2 percent from the nominal circuit voltage. The voltage drop measured at the line terminals of the control due to the x-ray equipment maximum load shall not exceed 3 percent.



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The following table indicates the power supply requirements for the various types of high-tension transformers based on the use of a single x-ray machine. The manufacturer should be consulted when more than one x-ray machine will be installed, or *if the run is longer than indicated in table I.

TABLE I. Electrical utility requirements

| High-tension transformer type (see 1.2.1.1) | Nominal line voltage | Distribution transformer capacity (KVA) | Line wire size (AWG) From | | X-Ray line disconnect switch | | |
|---|---|---|---|---|---|---|---|
| | | | Distribution transformer to disconnect *(for 100 ft.) | Disconnect to X-ray control *(for 25 ft.) | Ground wire size | Switch rating (amps.) | Fused for (amps.) |
| I | 240 | 7.5 | 6 | Supplied with equipment | 10 | 60 | 40 |
| II | 240 | 15 | 1 | | 10 | 60 | 60 |
| III | 240 | 37.5 | 0 | 4 | 8 | 100 | 100 |
| IV | 240 | 37.5 | 0 | 4 | 8 | 100 | 100 |
| V | 240 | 50 | 0000 | 2 | 6 | 200 | 150 |
| VI | 240 | 50 | 0000 | 2 | 6 | 200 | 150 |
| VII | 480 | 100 | 00 | 2 | 8 | 200 | 125 |

The above power supply and wiring shall be installed by the user prior to the installation of the x-ray equipment (see 6.2). Controls for types I and II high-tension transformers shall be supplied with line cables for connection from the x-ray line disconnect switch to the control and from the x-ray control to the high-tension transformer. These cables shall be of heavy duty cord with the appropriate number and sizes of conductors. Controls for types III, IV, V, VI, and VII high-tension transformers shall have provision for the direct connection of armored cable or conduit wiring supplied by the user in accordance with table I.

3.3.3.1.3.1 Supplier responsibility. If during the installation and calibration of the x-ray generator, the equipment supplier finds power supply conditions that could impair the efficient operation, reduce the normal life expectancy, or prevent the equipment from meeting any of the requirements of the specification, he shall immediately notify the contracting officer of the deficiencies and request instructions.

3.3.3.1.4 Line compensator. The line compensator incorporates all wiring, circuitry, and switching necessary to match the incoming line to the autotransformer in establishing the volt per turn ratio required for efficient and safe operation of all circuits fed by the autotransformer. This matching shall be accomplished within at least 1 percent accuracy. A line compensator device, operable from the control panel without the aid of tools, together with a meter for determining the proper adjustment of this device, shall be mounted in all control cabinets not employing a prereading type KVP meter which responds directly to line variations. In any event, components that may be adversely affected by line variations shall be adequately protected by stabilizing circuitry.

3.3.3.1.5 Milliampere technique selector. Each control shall have a milliampere technique selector providing a multiple choice of tube currents to be applied to an x-ray tube. These stations shall be electrically adjustable, to provide for exactness in calibration, to compensate for x-ray tube filament evaporation, and difference in x-ray tube filament characteristics.

3.3.3.1.5.1 Fluoroscopic techniques. The fluoroscopic and spotfilm station or stations of the MA technique selector shall provide for the application of fluoroscopic techniques to the fluoroscopic x-ray tube and, in conjunction with the operation of the spotfilm change-over switch, shall provide for the application to the proper spotfilm radiographic technique to the fluoroscopic tube. One spotfilm position shall be set for 200 MA but provision shall be made for adjusting this technique to any other value within the operational limitations of the focal spot of the fluoroscopic x-ray tube or x-ray generator at the discretion of the radiologist.

3.3.3.1.5.1.1 Fluoroscopic MA control. In conjunction with the fluoroscopic and spotfilm station of the MA technique selector, there shall be a fluoroscopic control by which the fluoroscopic milliamperage may be controlled from 1/2 MA up to, but not to exceed 5 MA. This control may be located on the control panel, but if located elsewhere, it shall be readily accessible to and adjustable by the operator without the employment of tools. Provision shall be made so that in the event an image intensifier adaption unit is supplied, the parts for automatically switching from the higher fluoroscopic MA to the lower for image intensification can be added without difficulty.



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3.3.3.1.5.2 Functions of MA technique selector. Adjustment of the milliampere technique selector shall accomplish the following functions:

(a) Provide for the proper energizing of desired filament of the x-ray tube under steady state conditions to insure the selected tube current when the exposure is made. Circuits shall be arranged so as to prevent excessive evaporation of the filament of the x-ray tube.
(b) Provide that the radiographic kilovoltage indicating device indicates within the prescribed accuracy the KVP that will be applied to the x-ray tube when the exposure is made.
(c) Select the proper kilovolt indicator lead compensation circuitry.
(d) Select the proper scale of the milliammeter.
(e) Compensate for the "space charge effect" for the MA techniques selected in coordination with the kilovoltage switch or switches, or by MA stabilization so that for any radiographic technique, the MA load remains constant within plus or minus 5 percent for the values of kilovoltage stipulated in 4.4.3.10.
(f) Select the proper exposure timing medium, e.g., the foot switch or radiographic timer.
(g) Provide for the coordination of other electrical circuits with the MA techniques selected as may be required in the operation of other allied apparatus.
(h) Control the high-tension switch and rotor starting circuits to select the radiographic or fluoroscopic x-ray tube.

3.3.3.1.6 Radiographic kilovoltage control or controls. Kilvoltage control or controls shall provide for a selection of the kilovoltage to be applied to the x-ray tube in steps of 3.0 KVP or less for types I and II high-tension transformers, or in steps of 2 KVP or less for types III, IV, V, VI, and VII high-tension transformers within the range of 50 KVP to maximum momentary rating of equipment specified in the invitation for bids. The control or controls shall provide for position indexing so that full contact is made without the possibility of making partial contact. The design shall be such that no contact filings will be able to accumulate between any two successive positions and thereby cause short circuits. When specified in the invitation for bid, the lower range of the radiographic kilovoltage control for types III, IV, V, VI, and VII high-tension transformers shall be extended downward to at least 26 KVP in increments suitable for mammography technique (see 6.1 (aa)).

3.3.3.1.6.1 Space charge effect. Provisions shall be made for coordinating the compensation of the "space charge" at the various settings of the kilovoltage control or controls with the MA techniques that may be selected with the MA technique selector switch. When tested as described in 4.4.3.10, the readings of the MA or MAS meter shall remain constant within 5 percent of the initial test setting.

3.3.3.1.7 Prereading kilovoltage indicator. The prereading kilovoltage indicator shall accurately indicate within plus or minus 3 KVP of the potential to be supplied to the x-ray tube for the MA technique selected. When a meter is supplied for prereading kilivoltage indication, its scale shall be at least 3-1/2 inches in length and it shall be accurate at 72° F. to within plus or minus 2 percent of full scale deflection for the activated scale (see 4.4.3.9).

3.3.3.1.8 Milliammeter. Each control shall incorporate a milliammeter whose scale shall be at least 3-1/2 inches in length. When tested as described in 4.4.3.6, the meter shall be accurate at 72° F. to within plus or minus 5 percent of full scale deflection for the activated scale. This meter shall be connected to coordinate the selection of scales with the MA technique selector, and MAS meter, when one is specified in the invitation for bid (see 4.4.3.6).

3.3.3.1.9 Radiographic timer. The control panel shall include a radiographic timer or timers for controlling either the duration of a radiographic exposure or quantity of MAS. Facilities shall provide for initiating the exposure at the discretion of the operator, either automatically after the rotor of the x-ray tube reaches safe rotative speed or at any instant after the rotor reaches safe rotative speed (see 4.4.3.5).

3.3.3.1.10 Tube indicator. When apparatus specified in the invitation for bid is to supply energy to more than one x-ray tube, a means shall be provided to indicate which tube is to be energized.

3.3.3.1.11 X-Ray tube rotor starting device. The rotor starting device shall provide for a safe rotational speed of the x-ray tube rotor to permit any exposure within the rating of the apparatus described in the invitation for bid. Provision shall be made to prevent an exposure should an open circuit occur in one or more leads to the stator circuit of the rotating anode to be energized. The starting device shall provide for the following:

(a) For fluoroscopic techniques, the stator shall be energized initially at sufficiently high voltage when the footswitch is pressed. After a time delay, the voltage shall be reduced to afford cooler operation and maintain adequate speed.
(b) For radiographic techniques with classes 1 and 2, style (a) rotating anode tubes, the starting device shall provide for exposure after the time delay. This shall be accomplished by



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initially energizing the stator with 220 to 230 volts; then after a time delay, reducing the voltage to 115 volts or less.

(c) Controls to function with types V, VI, and VII high-tension transformers shall be provided with rotor speed changing equipment to permit the operation of class 2, style (b) high speed rotating anode tubes at the ratings shown in 3.7.5.2 within a 2-1/2 second time delay. The rotor speed changing apparatus shall be mounted in the control cabinet or at a convenient external location. It shall be designed to operate from the 60 cycle single phase power supply available at the control cabinet. The speed changing equipment shall be connected automatically when the combination MA, KVP, and time exceeds tube limitations, or by a dual x-ray protective device on controls for use with type VII transformers (see 3.3.3.2.6). An automatic means shall be provided for braking the high speed rotor after an exposure (see 3.7.5.3).

3.3.3.1.12 Foot switch. Each control cabinet shall be equipped with foot switch circuits so that fluoroscopic techniques may be applied to the x-ray tube when the circuits of the control cabinet are set to deliver fluoroscopic techniques. Whenever type II, classes 2, 3, 4, and 5 or type III, classes 1 and 2 tilt tables are furnished as part of the x-ray apparatus, a foot switch of the dead-man type shall be supplied. This foot switch shall, in addition to having provisions for controlling the fluoroscopic x-ray exposure, be capable at the operator's discretion, of controlling whether or not the room light will flash "on" or "off" with the x-ray contactor or "on" and "off" without the operation of the contactor. When a spotfilm change-over device is incorporated with equipment, it shall be possible to make the spotfilm exposure either by depressing the foot switch or by any other exposure switch, that may be located on the spotfilm tunnel.

3.3.3.1.13 MA stabilizer. Each control cabinet shall be provided with a stabilizer which will maintain the x-ray tube current at the MA value selected by the MA technique selector within the limits described elsewhere in this specification for the various types of controls to function with the several different types of high-tension transformers (see 4.4.3.16).

3.3.3.1.14 Bucky switching. The bucky shall be connected in the radiographic position and disconnected in the fluoroscopic and spotfilm positions of the technique selector switch. If, as specified in the invitation for bid, more than one bucky diaphragm will be used, a selector switch shall be provided to connect and indicate which bucky is effective. When the milliampere technique selector is in the radiographic position, it shall be possible to make grid-free radiographs at exposure times of 1/10 of a second or longer with a parspeed bucky, or at exposure time of 1/30 of a second or longer with a superspeed bucky with the timing of the exposure controlled by the radiographic timer, or with the selected phototimer when included in the invitation for bid (see 6.1 (g)).

3.3.3.2 Additional control cabinet components. When specified in the invitation for bid, the control cabinet shall have any or all of the following components (see 6.1 (i)):

(a) Separate control for fluoroscopic kilovoltages.
(b) MAS meter or impulse MA meter.
(c) Fluoroscopic timer.
(d) Phototimer.
(e) Spotfilm change-over device.
(f) Capability of more than six exposures per second for use with types III, IV, V, VI, and VII high-tension transformers. (Rate desired shall be specified in the invitation for bid.)

3.3.3.2.1 Fluoroscopic kilovoltage control. The independent control of the fluoroscopic kilovoltage shall permit adjustment in increments of progressively increasing fluoroscopic screen brightness in no less than 10 steps from 65 to the maximum fluoroscopic ratings for controls for use with type III transformers, or as limited by the KVP rating of the transformer. Either a separate direct reading KVP indicator having an accuracy of plus or minus 3 percent or the same medium used for indicating the radiographic kilovoltage shall be used for indicating the fluoroscopic kilovoltage. Control shall be provided on the control cabinet or at a convenient location on the tilt table.

3.3.3.2.1.1 Automatic fluoroscopic KVP: When automatic brightness control is specified in the invitation for bid, the KVP shall automatically vary to maintain a constant image brightness while scanning over different body densities. There shall be set limits to the maximum range of this variable consistent with maximum 10 R/min. tabletop dose. Further provisions shall be made to provide this feature selectable, "in" or "out", at the discretion of the operator.

3.3.3.2.1.2 Remote fluoroscopic KVP: When specified in the invitation for bid a remote fluoroscopic KVP control shall be conveniently located for operation by the fluoroscopist. It shall cover the same range of KVP values as that located on the control cabinet.

3.3.3.2.2 Short exposure MA indication. An impulse MA meter or a MAS meter shall be supplied.

a. The impulse MA meter shall indicate average x-ray tube current for any exposure time increment from 1/120 second to the full time capability of the x-ray timer. The reading shall be accurate at any point of the scale to plus or minus 2 percent of the tube's actual current. The meter scale



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shall be not less than 3-1/2 inches long and shall be calibrated for fluoroscopy from 1/2 MA to not more than 5 MA on the fluoroscopic scale, and from not more than 10 MA to the maximum capacity of the generator on the radiographic scale or scales. The full value display of the tube current reading shall be held for a minimum of 4 seconds regardless of actual exposure time. In cineradiography, it shall read the average MA applied to the x-ray tube. Upon release of the x-ray exposure switch the meter activating current shall be totally discharged to provide for repeated exposures with the same degree of accuracy.

b. The MAS meter shall have a range of at least 0 to 50 and not more than 0 to 100 MAS. The length of the scale shall be not less than 3-1/2 inches and have calibrated divisions not greater than 2 MAS. The MAS meter may be incorporated in the same housing as the milliammeter and shall be for check purposes and only in the circuit when the operator pushes a switch to place it in the circuit. The accuracy of the MAS meter shall be within plus or minus 5 percent of full scale deflection for the activated scale for all exposure increments up to 3/10 of a second. Provision shall be made to prevent damage to the MA stabilizer in relation to the use of the MAS meter when electronic stabilization is employed (see 4.4.3.7).

3.3.3.2.3 Fluoroscopic timer. The fluoroscopic timer shall be of the synchronous motor-driven type. It shall integrate the intervals of time that the fluoroscopic tube is energized with high voltage and prevent further fluoroscopic examination after a preset time has expired, until the timer is reset. The scale of this timer shall be 0 to 5 minutes. The timer, unless otherwise specified in the invitation for bid, shall be located in the control cabinet of the x-ray apparatus.

3.3.3.2.4 Phototimer. When specified in the invitation for bid, a phototimer shall be included. This phototimer shall function only with the phototube receiver of the spotfilm tunnel when only one receiver is specified. When a total of more than one but not more than three phototube receivers located as below are specified in the invitation for bid, circuitry shall be provided for the proper operation of these receivers. The phototimer shall be tested for performance as described in 4.4.3.17 and shall meet the performance requirements listed in the following subparagraphs, as well as those in 4.4.3.17 (see 6.1 (j)).

(a) With spotfilm tunnel.
(b) With cassette changer.
(c) With photoroentgen equipment.
(d) With reciprocating bucky in x-ray table.

3.3.3.2.4.1 Phototimer switching. Switching means shall provide for either the radiographic timer or phototimer to control the x-ray exposure. Further, this switching means shall indicate which phototube receiver and aperture are effective. The functions of MA technique selection required for phototiming may be included in this switching means of by the MA technique selector of the control.

3.3.3.2.4.2 Reproducibility. The phototimer within the control and phototube receiver shall be provided with the necessary adjustments so as to produce equivalent quality radiographs of the same object from one time to the next. The critical electrical circuits of the phototimer shall be supplied with a stabilized voltage so that the normal line voltage fluctuations will not result in MAS varying more than 5 percent from one radiograph to another, when tested as specified in 4.4.3.17 (b).

3.3.3.2.4.3 Safety device. A safety device or circuit shall be provided to terminate the phototimed exposure prior to exceeding the safe time limit of the operating x-ray tube for the particular MA setting of each of the phototimer selector positions.

3.3.3.2.4.4 Density control. The density control shall provide adjustment to produce radiographs of the desired density. In addition to the coarse, or serviceman's adjustments, there shall be at least two (2) fine adjustments as described below in (a) and (b).

(a) One fine adjustment shall be located conveniently on the spotfilm device, designed for use by the radiologist without the use of tools, to provide him the choice of at least three film densities; namely, light, medium, or dark; for the films exposed in the spotfilm tunnel. This control shall, when set for maximum darkness, produce an MAS meter reading during exposure using a phantom equivalent to average patient density, of no greater than five times and no less than three times that produced when set for minimum darkness; technical factors within the average range of 80 KVP and 200 MA. The control shall be plainly marked to indicate the direction of movement to increase film darkness.

(b) One fine adjustment shall be located conveniently on the x-ray control, designed for use by the operator without the use of tools, to provide him the choice of at least three film densities namely, light, medium, or dark; for films exposed in the table bucky tray or cassette changer. This control shall, when set for maximum darkness, produce an MAS meter reading during exposure using a phantom equivalent to average patient density, of no greater than five times and no less than three times that produced when set for minimum darkness; technical factors within the average range of 70 KVP and 300 MA. The control shall be plainly marked to indicate the direction of movement to increase



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film darkness. This control shall provide coadaptation between the table bucky and the cassette changer phototiming stations obviating readjustment with each change of station selection.

3.3.3.2.5 Spotfilm change-over devices. The spotfilm change-over device shall be mounted in the x-ray control cabinet and incorporate the necessary spotfilm change-over circuits to function with the spotfilm tunnel to the x-ray tilt tables.

3.3.3.2.5.1 Functions. The spotfilm change-over device shall provide for the proper functioning and coordinated action of the following component parts of the equipment:

(a) X-ray tube focal spot selection.
(b) Rotor of rotating anode tube.
(c) Milliampere technique selector.
(d) Radiographic kilovoltage selectors.
(e) Fluoroscopic kilovoltage selectors.
(f) Kilovoltage indicating devices.
(g) Milliammeter.
(h) Radiographic timer circuit.
(i) Foot switch.

3.3.3.2.6 X-ray tube protective circuits. This apparatus shall be supplied for types III, IV, V, VI, and VII high-tension transformers. The control cabinet shall be provided with the tube protective circuits which shall integrate KVP, MA and time to prevent overloading the x-ray tubes. These circuits shall protect both the radiographic and fluoroscopic tubes. A signal light shall indicate the operation of the circuits and prevent an exposure that exceeds the tube rating. The control for a type VII transformer shall have dual protection to protect the tube when operating at either standard speed or high speed. A protective device adjusted for the high speed rating limits shall be supplied with the control for types V and VI transformers and a protective device for the standard speed rating limits shall be supplied with the control for types III and IV transformers.

3.3.4 Additional requirements for control cabinets to function with types I and II high-tension transformers.

3.3.4.1 MA techniques. Each control shall have a milliampere technique selector providing for at least four stations of fixed milliampere technique plus a fluoroscopic and spotfilm position. The four stations of fixed milliampere shall be as follows unless otherwise stipulated in the invitation for bid (see 6.1 (k)):

TABLE II. Milliampere stations

| Type I transformer | | Type II transformer |
|---|---|---|
| Small focus . . . . . | 25 MA to 50 MA | 25 MA to 50 MA |
| Large focus . . . . . | 75 MA to 100 MA | 100 MA to 200 MA |

3.3.4.2 Milliammeter. The scales of the milliammeter shall be at least 0 to 10 and 0 to 100 MA for type I transformers. The scales for use with type II transformers shall be at least 0 to 20 and 0 to 200 MA.

3.3.4.3 Radiographic timer. If an interval timer is supplied, it shall provide for exposures of 1/10 to at least 5 seconds for use with type I transformers. For type II transformers, the timer shall provide for exposures of 1/60 to at least 5 seconds. Either timer shall provide increments of approximate geometric progression. Within the range of 1/60 to 1/5 second, all exposures shall be of an even number of impulses and shall be correct to within 1 impulse. Exposures from 1/5 to 1 second shall be accurate within plus or minus 2 impulses. Above 1 second, maximum error shall be plus or minus 5 percent of the exposure time selected. If an interval timer is not supplied, an MAS type of timer shall be furnished. It shall be calibrated in at least 12 values of approximate geometric progression from 10 to 300 MAS. Tolerances for all values of MAS shall be plus or minus 5 percent (see 4.4.3.5).

3.3.4.3.1 Other timer requirements. Should the requirements of 3.3.4.3 not meet requirements of the user, these requirements shall be as listed in the invitation for bid (see 6.1 (l)).

3.3.4.4 X-ray contactor. The contactor shall carry the current of the primary of the high-tension transformer when an exposure is made and the main contacts shall be of nonfusing material to prevent "welding". Momentary surge protection at the initiation and termination of an exposure shall be provided. Initiation and termination of the exposure shall be accomplished within 1.5 milliseconds of zero voltage. If the contactor is used during fluoroscopy, it shall have contacts or a separate relay shall be furnished to disconnect the room light when the contactor closes. The contactor shall not be immersed in oil or other liquid.



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3.3.4.5 MA stabilization. Each control shall be provided with an MA stabilizer which will maintain the radiographic x-ray tube current at the MA value selected by the technique selector switch within 5 percent (see 4.4.3.16).

3.3.5 Additional requirements for controls to function with types III, IV, V, VI, and VII high-tension transformers.

3.3.5.1 MA technique. Each control shall have a milliamperage technique providing for at least seven stations of fixed milliamperage technique plus at least one fluoroscopic and spotfilm position. The stations of fixed milliamperes shall be as follows unless otherwise stipulated in the invitation for bid (see 6.1 (m)).

TABLE III. Milliampere stations

| | Types III and IV transformers | Types V and VI transformers | Type VII transformer |
|---|---|---|---|
| Small focus | 25, 50, 100 | 25, 50, 100 | 25, 50, 100 |
| Large focus | 100, 150, 200, 300 | 200, 300, 400, 500 | 200, 300, 400, 500, 600, 800 |

3.3.5.1.1 MA techniques for more than one radiographic tube. In apparatus employing classes 2 and 3 high-tension transformers with two or more radiographic tubes, each radiographic technique selector position shall be made available for each tube. The actual MA techniques of the selected x-ray tube shall be within 10 percent of the value indicated on the MA technique selector.

3.3.5.2 Milliammeter. A milliammeter having double or triple scales, shall be supplied with each control. The calibration of each shall provide for easy unmistakable readings. The necessary apparatus and circuits shall be supplied to provide correct MA indications of the actual MA output within 5 percent accuracy when tested as described in 4.4.3.6.

3.3.5.3 Radiographic timer range. The timer or timers shall provide for exposures of 1/120 second to a maximum of 5 to 8 seconds duration in increments of approximate geometric progression. Within the range of 1/120 to 1/5 of a second, all exposures shall be correct to within complete impulses. An impulse is that portion of the sine wave starting within 32 and continuing to at least 148 degrees. Within the range of 1/5 to 1 second, the exposure shall be correct to within 1 impulse. Within the range of 1 second to maximum scale of the timer, the exposure shall be correct to within 2 percent of the exposure setting selected. Accuracy shall be determined as described in 4.4.3.5.

3.3.5.3.1 Timer recycling. Provision shall be made for recycling the timer to permit the repetition of 1/20 of a second exposures at the rate of 6 exposures per second. The 6 1/20 second exposures shall be accomplished when used with rapid film changing equipment where 60 percent of the time is utilized in film transport (see 4.4.3.11).

3.3.5.3.2 Other timer requirements. Should a timer having characteristics other than those specified herein be required, such characteristics shall be stated in the invitation for bid (see 6.1 (n)).

3.3.5.4 MA stabilization. Each control shall maintain the radiographic x-ray tube current at the MA value selected by the MA selector within 5 percent throughout the timing range (see 4.4.3.16).

3.3.5.5 X-ray contactor. Either electromechanical or electronic type contactors providing the following shall be acceptable:

(a) Capable of initiating and terminating the exposure within 1.5 milliseconds of zero voltage.
(b) Capable of producing in excess of six separate expsoures per second.
(c) Provide for surge protection and generate no transient peak-to-peak spike wave configurations greater than 15 percent of the peak pulse under observation as representing the KVP applied to the x-ray tube.
(d) It shall carry the current of the primary to the high voltage transformer with safety provisions for at least 100 percent overload based on the maximum load for the H.V. transformer to the maximum rating of the Hi-speed x-ray tube.
(e) Mechanical contacts shall be fabricated of nonfusing material to prevent welding.
(f) If used for fluoroscopy, it shall provide circuitry for room-light disconnect during exposure.
(g) It shall not be immersed in oil or other liquid.
(h) It shall provide for component replacement of parts rather than assembly replacement.

3.4 Tube stands.

3.4.1 Common requirements.



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3.4.1.1 X-ray tube accommodation. The tube stand shall be capable of properly accommodating an x-ray tube and allied components having a combined weight not exceeding 80 pounds, and shall be counterbalanced for the attached load.

3.4.1.2 X-ray tube positioning. Each tube stand shall provide for the easy adjustment and positioning of the x-ray tube as may be required to function with any of the equipment described in the specification.

3.4.1.2.1 X-ray tube tabletop centering. It shall be possible to position the focal spot of the radiographic x-ray tube over the radiographic table area so as to permit a normal central ray to be centered within 6 inches of the lengthwise edges of the tabletop as measured in the horizontal plane.

3.4.1.2.2 Lateral x-ray tube position. With the tube carriage of the tube stand positioned to send a ray directly across the narrow dimension of the tabletop for lateral radiography, it shall be possible to bring the central ray of the x-ray tube within 1-1/2 inches of and parallel to the horizontal tabletop.

3.4.1.2.3 X-ray tube position for vertical bucky radiography. With the tabletop positioned for vertical bucky radiography, it shall be possible to position the tube stand to obtain a bucky film-to-target distance of 40 inches with a tabletop coverage as indicated in 3.4.1.2.1. This shall apply to type III tables when in either the 88° Trendelenburg or vertical position.

3.4.1.2.4 Longitudinal positioning of the x-ray tube. The longitudinal movement of the tube stand shall provide for positioning the x-ray tube in such a manner as to direct the central ray normal to the floor within the limitations as described in 3.4.1.2.1 plus the ability to angle the x-ray tube 45° toward the foot end of a horizontal table with 36 inch target-to-film distance and have the central ray strike the center of the cassette centered in the cassette tray of a bucky positioned as far as possible to the head end of a table in the horizontal position.

3.4.1.3 Tube carriage.

3.4.1.3.1 Rigidity of tube carriage. The tube carriage shall be rigidly designed for ease and convenience of x-ray tube manipulations and have no sag greater than 1/4 inch from the horizontal for any position of the tube carriage and x-ray tube.

3.4.1.3.2 Rotation of tube carriage about its vertical support. The tube carriage upon its vertical support shall permit all radiographic procedures described in 3.4.1.1 through 3.4.1.2.4. The tube carriage when rotated about its vertical support shall have mechanical indexing positioning at each 90° through plus and minus 180° when measured from the normal radiographic position during tabletop radiography.

3.4.1.3.3 Transverse stereoscopic shift. A means shall be provided to permit transverse stereoscopic shifts up to at least 4 inches with distance indicators.

3.4.1.3.4 Longitudinal stereoscopic shift. A means shall be provided to permit longitudinal stereoscopic shifts up to at least 4 inches with distance indicators.

3.4.1.3.5 Vertical stereoscopic shift. A means shall be provided to permit electrically controlled vertical stereoscopic shifts over a range of at least 3.6 to 7 inches. This shift shall be either directly adjustable or preset to a fixed value within this range. The shift shall be controlled in such a manner that the x-ray tube shall be brought to rest and all vibration reduced to less than 0.25 mm. within 3 seconds after the shift is started.

3.4.1.3.6 Alignment of central ray. A means shall be provided with an indicator, or by an electrically operated lock for aligning the central ray of the x-ray tube to the center of the horizontal tabletop with respect to its width.

3.4.1.3.7 Angulation of central ray in the lengthwise plane normal to the horizontal tabletop. With respect to a central ray normal to the horizontal tabletop, provision shall be made for a plus and minus angulation of 180° (360° total) of the central ray. The axis of rotation of this angulation shall be about a line parallel to the width of the horizontal table and through the target of the x-ray tube.

3.4.1.3.8 Provision for angulating the central ray in the transverse plane normal to the horizontal tabletop. With respect to a central ray normal to the horizontal tabletop, provisions shall be made for an angulation of not less than 30° forward and 20° backward, a total angulation of not less than 50° transversely in the plane normal to the horizontal tabletop. The axis of rotation shall be approximately through the center of gravity of the tube carriage assembly with adjustable diaphragm collimator. This angulation shall be possible when the radiographic collimator is used. Motion shall be smooth and free of whip or backlash.



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3.4.1.3.9 Radiographic filter. The total filtration, permanently in the useful beam, shall not be less than 2.5 mm. of aluminum equivalent. Normally, a 2.0 mm. aluminum filter shall be fixed in the useful beam which can only be removed by the use of common tools. (The filter shall not be in the port of x-ray tube.)

3.4.1.3.10 Scales and locks. All movements and angulations of the tube stand, tube carriage, and x-ray tube shall be provided with locks which operate easily and are convenient to the operator. Except where otherwise specified, electrically operated locks are desirable, but not mandatory as long as a technician 5 ft. 3 inches tall can easily operate the locks. All angulations affecting the position of the x-ray tube shall be provided with scales that are easily read and permit accurate positioning.

3.4.1.3.11 Radiographic collimator. An adjustable multidiaphragm collimator shall be provided with a means for easy adjustment of each set of diaphragms independently. The diaphragms shall be spaced as far apart as practical so that the exposed radiograph shall be free of penumbral shadows produced by scattered radiation from the x-ray tube, tube housing, cone, etc. The adjustments of the diaphragms shall be simple and shall be provided with direct readings calibrated for coverage of various size x-ray films ranging from 6 to 17 inches at 36, 40, 48, and 72 inch target distances. A light with "ON" and "OFF" interval-timed switch shall illuminate the area to be irradiated with an indication of the central ray and means for centering with the bucky. The inherent filtration of the collimator shall be approximately equivalent equivalent to 1.0 mm. aluminum. The central ray from the targent shall coincide with the center of the collimator with the same tolerance for both focal spots as permitted for the x-ray tube. The fabrication of the multidiaphragms shall be such that screws will not become loose with normal operation of the diaphragms.. The collimator shall be designed for radiography at 150 KVP.

3.4.1.3.11.1 Multidiaphragm collimator defined. A multidiaphragmed collimator is one employing more than one radiation restricting diaphragm. The arrangement of the diaphragm shall be such that secondary radiations generated within the collimator are absorbed within the collimator.

3.4.2 Additional requirements for type I tube stands.

3.4.2.1 Tube-stand support. The entire weight of the overhead mounted tube-stand shall be supported from the side walls and ceiling.

3.4.2.2 Ceiling height. The type I tube-stand shall be constructed to fit into a room whose ceiling height is not less than 9 nor more than 10-1/2 ft. (see 6.1 (o)).

3.4.2.3 Additional radiographic area. The design shall be such as to permit radiography of a patient on a wheeled stretcher when placed adjacent to the x-ray table. A transverse travel of at least 6 ft. shall be provided. Other criteria describing the radiographic area to be covered shall be as specified in the invitation for bid. Requirements of 3.4.1.2.2 shall be accomplished from front and back sides of the table (see 6.1 (p)).

3.4.2.4 Vertical positioning of the x-ray tube. The target-to-horizontal tabletop distance shall be vertically variable up to a maximum of not less than 48 inches. The tube-stands shall be counter-balanced to provide smooth adjustment averaging no more than 7 pounds effort to effect vertical movement of tube-head and collimator.

3.4.2.5 Electric locks. The longitudinal and lateral motions of the x-ray tube shall be provided with electrically-operated locks. The locks shall be fast acting, and not unreasonably loud.

3.4.3 Additional requirements for type II tube-stands.

3.4.3.1 Ceiling height. The floor-to-overhead supported tube stand shall be so constructed to be accommodated by any ceiling height from 9 to 12 ft. The target-to-horizontal tabletop film distance shall be vertically variable up to a maximum of not less than 48 inches (see 6.1 (q)).

3.4.3.2 Interference with tilt table. When the radiographic tube-stand is placed in the "parked position", the tube-stand, tube, or cables shall not interfere with the tilting of the x-ray table, its spotfilm device, or fluoroscopic tower.

3.4.3.3 Floor rail. The floor rail for this tube-stand shall be such that when installed, a wheel stretcher can pass over it.

3.4.3.4 Overhead support. All materials shall be supplied for fastening the overhead support to the ceiling or walls.

3.4.3.5 Length of floor rails and overhead support. Sufficient length shall be provided to permit 40 inches T. F. D. to vertical bucky in the tilt table or to permit AP and lateral radiography on mobile 24 by 24 inches apparatus positioned at head end of x-ray table when in horizontal position.



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3.5 Tables.

3.5.1 Construction. Each table shall be constructed of materials of sufficient strength to support a 300-pound patient.

3.5.2 Tabletops. Each tabletop shall be constructed of such material that its opacity to x-radiation will be not greater than 1.0 mm. of aluminum equivalent when measured at 80 kilovolts without a patient or phantom. With a power driven tabletop, the total opacity to x-radiation including the cover shall not be greater than 1.5 mm. aluminum equivalent measured at 80 kilovolts without a patient or phantom. The opacity shall be uniform for its entire area. Under fluoroscopic examination, the tabletop shall not show any inclusions or variation in density. The tabletop assembly and support frame shall be of such rigidity that a 300-pound weight concentrated in an area of 1 sq. ft. anywhere on the top will not cause the top to sag, which will result in an effort greater than 8 pounds to move the bucky along the tracks.

3.5.2.1 Tabletop marking. The tabletop, except style (c), shall be inscribed with clearly visible lines as follows:

(a) One line through the center of the tabletop running parallel to the length of the table.
(b) One line across the width of the tabletop indicating the center of the bucky diaphragm grid when the bucky is moved as far as possible to the head end of the table (tabletop in its normal centered position).
(c) One line across the width of the tabletop indicating the center of the bucky diaphragm grid when the bucky is moved as far as possible to the foot end of the table (tabletop in its normal centered position).

NOTE: These lines shall not be visible in films exposed in the bucky during radiographic procedures or upon the fluoroscopic screen during fluoroscopic procedures.

3.5.2.2 Binding strips. The tabletop shall not be fastened to the table with exposed screws, raised binding, or finishing strips that would retain dirt or barium.

3.5.3 Mounting device for accessories. Each table shall be provided with provisions for mounting detachable accessories. These provisions shall be designed to readily permit ease in cleaning. Their construction shall be such as not to retain barium.

3.5.3.1 Accessories. The following accessories shall be supplied if specified in the invitation for bid (see 6.1 (r)):

(a) Head clamp.
(b) Shoulder support (not required for type I tables).
(c) Compression device.
(d) Footrest (not required for type I tables).
(e) Radiation shield for type II, classes 2, 3, 4, 5, 6 and type III tables.
(f) Mounting step for types II and III tables.
(g) A substitution of parspeed bucky diaphragm in lieu of super-speed type.
(h) Two way power-driven tabletop for types II and III tables.
(i) Four way movable tabletop for type III tables. (Longitudinal travel by power-lateral travel by power or manual.)
(j) Myelographic patient support for type III tables.

3.5.3.1.1 Head clamp. The head clamp shall be the self-centering type and shall be (a) capable of being attached to the head end of the table, (b) flexible enough for any skull positioning, (c) rigid enought to provide immobilization, (d) comfortable to the patient, and shall be such that it can be (e) easily and quickly applied, and (f) easily kept clean. Where moving tabletops are specified, this clamp shall move with the top.

3.5.3.1.1.1 Longitudinal adjustment. The head clamp shall be capable of extending at least 17 inches over the tabletop from the head end, adjustable through a range of at least 7 inches, and shall be provided with a positive hand operated lock or locks.

3.5.3.1.1.2 Vertical adjustment. The head clamp shall be adjustable in its movement to and from the tabletop for a distance of at least 2-1/2 inches and be provided with a positive, hand operated lock for this movement. This movement shall cover the area from 2-1/2 to 5 inches above the table.

3.5.3.1.1.3 Fork adjustment. The unit shall be equipped with a clamp fork capable of being rotated 360° about its axis with the entire head unit capable of being displaced to either side of the midline to accommodate eccentric positions. The jaws of the clamp shall be capable of opening and closing through a range of at least 9 inches. The adjustment of the jaws of the head clamp shall be accomplished by a screw mechanism operated from the extreme head end of the table through a shaft.



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3.5.3.1.1.4 Resilient pads. The device shall be equipped with a resilient pad fastened to an arm which slides in and out of the clamp shaft for positioning against the top of a patient's head. The pads, touching the sides and top of the patient's head, shall be self-adjustable to the contour of the patient's head.

3.5.3.1.2 Shoulder support. The shoulder support shall be designed to support a 300-pound patient adequately, and shall be cushioned in proper places to provide comfort to the patient. The support shall not interfere with radiographic or fluoroscopic examinations.

3.5.3.1.3 Compression device. The compression device shall be designed so that it can be attached to the sides of the table. Three radiolucent bands (2 spares) of sufficient dimensions to be used across the abdomen of a 300-pound patient supine on the tabletop, shall be provided. It shall be possible to use this compression device at any point along the length of the table. A suitable ratchet or other type of mechanism shall be located on the operator's side of the table to provide for tightening of the band and shall be so designed that the compression may be easily released by the operator.

3.5.3.1.4 Footrest for types II and III tables. The footrest supplied shall be designed to support a 300-pound patient. The footrest shall be so designed that it can be fastened to the accessory mounting device and be adjustable over the upper and lower third of the tabletop length. When tabletop is of the movable type the footrest shall move with the top. It shall not be possible to mount the footrest so that it could strike the floor when the table is tilted.

3.5.3.1.5 Radiation shield for type II, classes 2, 3, 4, 5, 6 and type III tables. The radiation shield shall be mounted on the front side of the tables to help protect the operator from scattered radiation during fluoroscopy. A means shall be provided for convenient parking of this shield so as not to interfere with the use of the bucky. The shield shall project approximately 6 inches above the tabletop when in use. This shield shall have an equivalent lead protection of 0.25 mm. at 100 KVP.

3.5.3.1.6 Mounting step. The step shall be mounted on the front side of types II and III tables to assist the patient in mounting on the horizontal tabletop. The step shall be approximately 13 by 20 inches and covered with skid proof material. It shall be retractable when not in use.

3.5.3.1.7 Power-driven tabletop for type II, classes 2, 3, 4, 5, 6 and type III tables. The longitudinal movement of the power-driven top shall provide for coverage of an angiographic unit at least 18 inches wide by 31 inches high when it is located at the head or foot end of the table.

Style (b) Power-driven tabletop shall have a two-way motor drive for 30 inches longitudinal extension of the top beyond either end of the body, controlled by a switch on the fluroscopic assembly as well as switches located on the front side of the table.

Style (c) Power-driven tabletop shall have a four-way motor drive, or the lateral motion may be manually operated, for 30 inches longitudinal extension of top beyond either end of body, and for at least a transverse motion of 4.5 inches to the front and rear of the table centerline, for a total excursion of at least 9 inches.

The motor control switches shall be located on the fluoroscopic assembly and on the table body. The travel rate shall be nominally 2 inches per second longitudinally and 1 inch per second transversely, when power-driven. Shall be available for use on type III tables only.

3.5.3.1.7.1 Tabletop and cover. The power-driven top shall support a patient load of 300-pounds and shall move freely with that patient load. The table shall have a cover mounted in the table below the top to protect the interior parts when the top is extended.

3.5.3.1.7.2 Safety device. The power-driven table top shall be provided with a safety device to prevent tilting the table when the top is extended with insufficient ceiling clearance.

3.5.3.1.8 Myelographic patient support. When a table with 88° Trendelenburg tilt is specified in the invitation for bid, a myelographic patient support shall be supplied. This support shall consist of a pair of boots with fittings for attachment to the footrest or to a separate support that can be fastened to the accessory mounting device. The support shall safely support the weight of a 300-pound patient.

3.5.4 Power assist. When specified in the invitation for bid, a power assist apparatus shall be provided for the movement of the fluoroscopic assembly in the lengthwise direction of the table when type II classes 2, 4, 5, 6 and type III tables are specified. The power assist shall provide for the safe movement of the fluoroscopic assembly in the lengthwise direction of the table when a force not to exceed 3 pounds is applied by the operator. Adjustments shall provide for varying the speed or the force required for operation. The activating switch shall be incorporated within the handle of the spotfilm tunnel or image intensifier support. Provisions shall be made to provide smooth stops to prevent damage from weight inertia.



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3.5.5 Buckys. Unless a parspeed bucky is prescribed in the invitation for bid, each table shall be equipped with a superspeed reciprocating bucky diaphragm. The bucky shall be easily positioned and securely locked at any place along the full length of the table. The bucky shall be equipped for phototiming when a general radiographic phototimer with provisions for bucky radiography is specified in the invitation for bid. Buckys mounted in tilting tables shall be counterbalanced without a cassette but with a grid. The table and the bucky mounting shall be sufficiently rigid to prevent any vibration that could impair the quality of the bucky radiographs (see 6.1 (s)).

3.5.5.1 Timing range. When tested as described in 4.4.3.13, the superspeed bucky shall provide for automatic, continuous reciprocating motion of the grid to allow the product of reasonably grid-free radiographs at exposures from 1/30 of a second up to any exposure time required for bucky radiography when used with types III, IV, V, VI, or VII high-tension transformers. With the parspeed bucky, it shall be possible to make reasonably grid-free radiographs at exposures from 1/10 of a second up to any exposure time required for bucky radiography. The operation of the bucky shall be controlled from the hand exposure switch.

3.5.5.2 Timer starting switch. The reciprocating bucky shall be equipped with a switch actuated by the motion of the grid to start each exposure at the proper moment during the grid travel. The actual timing of the bucky radiograph shall be accomplished by the timer in the x-ray control cabinet.

3.5.5.3 Bucky grid. A removable grid with a 12 to 1 ratio shall be supplied with each bucky.

3.5.5.3.1 Grid construction. Each grid shall be composed of not less than 80 lead strips per inch, separated by radiolucent material. The radiolucent material and lead strips shall be securely held together by suitable facings plus water-proof coatings. Such construction and finish shall be sufficiently moisture-proof to resist warpage and separation. The lead strips shall be uniformly spaced with angulations to provide for effective centering at a 40-inch radius.

3.5.5.3.1.1 Grid radiographs. Radiographs of the grids shall be supplied with each bucky. The radiographs shall indicate that the grids are free from foreign objects, multiple lead or wood strips, strips not uniform, excessive lead turn-over, and areas of unequal density, as compared with the manufacturer's standardization films which shall be furnished upon request for comparison during test procedures (see 4.4.3.13).

3.5.5.4 Bucky trays. Each bucky shall be equipped with a tray having a cassette centering device accommodating both lengthwise and transversely, all sizes of standard cassettes from 8 by 10 to 14 by 17 inches, inclusive, complying with Fed. Std. No. 76. The tray shall be arranged so that once a cassette is centered in it, the centering device can be secured in position and readily be released by the operator.

3.5.5.5 Bucky centering device. An indicating device shall be provided for centering the bucky radiographically with the tube stand.

3.5.5.6 Bucky tabletop distances. The bucky shall be mounted in the table so that the distance from the top of the table to the film in the bucky tray shall be not more than 2-5/8 inches. This distance may be increased by 3/4 inch when the power-driven tabletop is specified.

3.5.5.7 Bucky wiring. The electrical wiring to the bucky shall be made by continuous electrical conductors with provisions for absorbing the slack in the bucky cable within the table proper. Sliding or moving contacts within the table will not be acceptable with regard to the bucky circuits.

3.5.5.8 Effective bucky tabletop coverage. Type I, class 1 tables shall have an effective bucky tabletop coverage of not less than 58 inches. Types II and III tables with stationary tops (style (a)), shall have a tabletop bucky coverage of at least 62 inches. Movable tabletops (styles (b) and (c)), may have the actual bucky travel reduced to a minimum of 54 inches, but the effective tabletop coverage shall be increased by the longitudinal travel provided by the movable top.

3.5.6 Nontilting radiographic tables.

3.5.6.1 Class 1 horizontal bucky table. The horizontal bucky table shall be open under the bucky and shall not tilt.

3.5.6.2 Class 2. Pedestal base table, for use in cardiac catheterization, without bucky.

3.5.6.2.1 Construction. The table shall be of the open type. The pedestal shall be centrally located to provide access to the table from all sides.

3.5.6.2.2 Tabletop travel. The tabletop shall be mounted on heavy duty bearings to fulfill the following conditions without any part of the table appearing on the radiographs. The following are minimum requirements:



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Longitudinal travel - 47 inches or more.
Lateral travel - 9 inches or more.
Overhang at head end - 50 inches.
Overhang at foot end - 50 inches.

The movement of the top shall require no more than 5 pounds of effort. It shall provide for the coverage of an angiographic unit approximately 18 inches wide by 31 inches high when it is located at the head or foot end of the table. Electrically operated locks shall be provided for all movements of the top. There shall be a centering device to indicate when the tabletop is in the center of its lateral travel.

3.5.7 Types II and III tables.

3.5.7.1 Construction. Types II and III tables shall be of the enclosed type, of a material suitable for retaining scatter radiation. The rear section shall be mounted to a floor supported base. There shall be no protruding front leg. The tables shall be accessible from all sides and be supported from the floor only.

3.5.7.1.1 Protection. A retractable device shall be provided to automatically cover the bucky slot opening at the front of the table when the bucky is "parked".

3.5.7.2 Table-tilting mechanism. The tables shall be equipped with a power tilting mechanism with a tilting range as specified in the invitation for bid (see 3.5.7.2.4).

3.5.7.2.1 Tilting speed and operation. The table tilting mechanism shall be designed to tilt the table quietly, smoothly, and without vibration and jerky starts and stops. Type II tables shall have at least one tilting speed at the rate of 4.5° plus or minus 0.3° per second. Type III, class 1 tables shall have continuously variable speed control at the rate of 1.2° to 4.5° per second, or at least 2 tilting speeds, 4.5° per second and another slower speed. Type III, classes 2 and 3 tables shall have continuously variable speed control at the above rates. The tilting speed shall be measured with a 300-pound patient lying on the tabletop. The tilting of the table shall decelerate smoothly to a complete stop in 2 seconds upon release of power to the tilting mechanism. It shall be possible to control the tilting of the table from the operator's fluoroscopic positions (horizontal and vertical). Hand-operated switches, if used for controlling the tilting, shall be such that they can be easily operated by an operator wearing protective gloves.

3.5.7.2.2 Automatic stops. Controls and limiting switches for the tilting table shall be designed to stop the table automatically in the maximum Trendelenburg, horizontal and vertical positions. A positive stop for the horizontal position shall be provided, or the tilting mechanism shall provide a hesitation in the horizontal position of sufficient duration that the operator can release the tilting switch when the tabletop is in the horizontal position. Under no condition shall it be possible to activate the "down" control and have the table tilt upward or activate the "up" control and have the table tilt downward.

3.5.7.2.3 Tilting indicator. A suitable and accurate scale, mounted in a readily visible position, shall indicate the angulation of the table with respect to the horizontal.

3.5.7.2.4 Tilting range (see 6.1 (t)). **The tilt range for the tables shall be as follows:**

Type II - Not less than 15° Trendelenburg to not more than 2° of vertical.
Type III - Not less than the degrees Trendelenburg specified in the invitation for bids (usually 90°) to not more than 2° of vertical.

3.5.7.3 Fluoroscopic assemblies.

3.5.7.3.1 Fluoroscopic x-ray tube mounting. The fluoroscopic x-ray tube shall be mounted on the fluoroscopic assembly of the table in such a manner as to fulfill the following conditions without the image of any part of the table appearing on the fluoroscopic screen:

Tube target tabletop distance . . . . . 18 inches or more.
Fluoroscopic tube cross travel. . . . . 10 inches or more.
Fluoroscopic tube lengthwise travel with fixed top. . . 40 inches or more.
Fluoroscopic tube lengthwise travel with power-driven top 36 inches or more.
Radiation cone to confine beam to useful area:

3.5.7.3.1.1 Fluoroscopic x-ray tube lengthwise travel. The lengthwise travel of the fluoroscopic x-ray tube shall be so arranged that with the table in the vertical position, the maximum focal-spot floor distance shall be not less than 66 inches. For tables having a Trendelenburg of 15° or less, the above distance shall be not less than 63 inches.



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3.5.7.3.1.2 Shutters. The fluoroscopic tube-mounting device shall be equipped with 2 pairs of shutters operating independently at right angles to provide closure in either direction. The shutters shall be positioned as close to the tabletop as practicable. The shutters shall operate smoothly and easily and shall be centered with the central ray of the fluoroscopic x-ray tube. The shutters shall have a minimum lead equivalent primary protection of 2.7 mm. of lead at 150 KVP. The opening of the shutters shall be continuously limited to restrict the maximum fluoroscopic field to within approximately 1/4 inch of the edges of the fluoroscopic screen or image intensifier shielding. In no position may a useful beam extend beyond the limits of the fluoroscopic screen or the shielding of the image intensifier. Type II, class 6 and type III, class 3 tables shall be equipped with circular type shutters that limit the beam to the face size of the image intensifier tube provided.

3.5.7.3.1.2.1 Shutter control. The shutter controls shall be located conveniently on the left-side of the spotfilm device with respect to the normal vertical position of the tilt table. The controls shall be easily operated by an operator wearing gloves having 0.5 mm. of lead equivalent protection. The shutter controls may be manual or motor driven and may be of the dual or single control type. Automatic programmed shutter controls shall be provided with type III, class 2 and type II, class 5 tables. Automatic programmed shutters shall provide automated spot radiograph size selection, full or fractional, maintaining a constant size exposure area regardless of distance relationship between spotfilm device and tabletop.

3.5.7.3.1.3 Added filtration. The total filtration permanently in the useful beam shall be not less than 2.5 mm. aluminum equivalent. Normally a 2 mm. aluminum filter shall be fixed in the beam path.

3.5.7.3.2 Fluoroscopic tower. The fluoroscopic tower shall be mechanically connected to the fluoroscopic tube mounting device and shall support the spotfilm tunnel and be adaptable to accept an image intensifier.

3.5.7.3.2.1 Tabletop screen distance. The fluoroscopic tower shall be so constructed that the spotfilm tunnel can be moved to and from the tabletop within the range of 7 to 18 inches to the underside of the spotfilm device. The spotfilm device shall be supported so that in its functioning position it is always in a plane parallel to the top of the table.

3.5.7.3.2.2 Fluoroscopic assembly counterbalancing. The fluoroscopic assembly with a cassette shall be counterbalanced in all functional positionings at all angles of elevation of the tilt table.

3.5.7.3.2.3 Creep. The fluoroscopic assembly shall be constructed so it will not creep longitudinally or laterally regardless of the elevation of the tilt table.

3.5.7.3.3 Motions. All motions of the fluoroscopic assembly shall be performed easily, firmly, and smoothly.

3.5.7.3.4 Locks. Locks, easy to operate and positive in action, shall be provided for all motions of the fluoroscopic assembly. These locks shall be convenient to the operator at any angle of elevation of the tilt table and capable of operation by an operator wearing protective gloves having 0.5 mm. of lead equivalent protection. These locks shall automatically engage when the spotfilm cassette tray is in other than its fluoroscopic position.

3.5.7.3.4.1 Myelographic stop. There shall be provided an adjustable stop which will prevent the fluoroscopic assembly from approaching closer to the tabletop than a predetermined distance and which shall not interfere with any other movement of the fluoroscopic assembly. This lock shall be suitable for use in myelographic procedures.

3.5.7.3.4.2 Compression lock. A lock shall be provided for the fluoroscopic assembly to retain compression during the fluoroscopic examination. This lock shall be provided with immediate release capabilities that are readily accessible to the operator.

3.5.7.3.5 Spotfilm device parking. The design of the fluoroscopic tower shall be such that the spotfilm device can be "parked" to permit radiographic procedures on the tabletop without interference when used with type I tube stands.

3.5.7.3.6 Interlock. The design of the "parking" mechanism for spotfilm device shall be sturdy and if, when in its parked position, it would interfere with the safe operation of the table, a switch interlock shall be provided to prevent tilting of the fluoroscopic table until the spotfilm device is placed in a safe operating position.

3.5.7.3.7 Spotfilm tunnel.

3.5.7.3.7.1 Spotfilm procedure. Fluoroscopic exposures shall be controlled by the foot switch. For fluoroscopic tubes having larger focal spot sizes than 0.7 mm. the rotor may be stationary



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during fluoroscopy. The equipment shall be such that when the operator causes the cassette to move from its x-ray protected position to its properly centered spotfilm radiographic position, the electrical circuits of the spotfilm change-over device in the control cabinet will cause the following action to take place automatically as soon as movement begins:

(a) Energize the fluoroscopic rotating anode tube to bring the rotor to the required safe speed.
(b) Properly energize the desired filament of the fluoroscopic tube for the spotfilm exposure.
(c) Change KVP from the fluoroscopic setting to the desired radiographic setting.
(d) Connect timing and foot switch circuits so that a timed exposure will be made by depressing the foot switch.

3.5.7.3.7.2 Spotfilm tunnel mounting. The spotfilm tunnel shall be permenently fastened to the fluoroscopic tower in such a manner that it can be "parked" to permit radiographic procedures, using the "over-the-table" x-ray tube without interference of the "parked" tunnel with the positioning of the tube stand or the patient. An interlock switch shall be provided to prevent an exposure through the fluoroscopic x-ray tube when the tunnel is in its "parked" position, out of the path of the central ray. The tunnel and its mounting shall be so designed that in any operating position, the fluoroscopic tube central ray shall be within 1/4 inch of the geometric center of the visible fluoroscopic screen and spotfilm exposures when the tunnel is positioned 15 inches from the tabletop.

3.5.7.3.7.2.1 Types of mounting: Hinged or vertical type parking shall be supplied with type II, classes 2, 3, 4, and type III, class 1 tables. Glide back or horizontal type parking shall be used with type II, class 5 and type III, class 2 tables.

3.5.7.3.7.3 Spotfilm tunnel classification. The spotfilm tunnel shall be of the following types, styles, and classes as specified in the invitation for bid (see 6.1 (u)).

Type I. Semiautomatic cassette movement - for use on type II, class 3 tables, with 8 by 10 inch cassettes.

Class (1) - Without phototiming.
Class (2) - With phototiming.

Type II. Automatic cassette movement.

Style (a) - For use with 8 by 10 inch cassettes - for use on type II, classes 2 and 4, and type III, class 1 tables.
Class (1) - Without phototiming.
Class (2) - With phototiming.

Style (b) - For use with 9-1/2 by 9-1/2 inch cassettes - for use on type II, class 5 and type III, class 2 tables.
Class (1) - Without phototiming.
Class (2) - With phototiming.

3.5.7.3.7.3.1 Image intensifier adapter. The spotfilm tunnel supplier shall make available upon demand, a selection of an adapter with design characteristics for adaptation of image intensifiers whose input phosphor is either a nominal 6 or 9 inch diameter.

3.5.7.3.7.4 Spotfilm combination. Spotfilm tunnels shall provide at least the following choice of exposures on a single film:

Style (a):

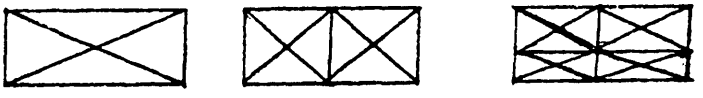
8 by 10 inch - long axis longitudinal with table:

1 exposure, 8 by 10
2 exposures, 4 by 10



8 by 10 inch - long axis transverse to table:

1 exposure, 8 by 10
2 exposures, 5 by 8
4 exposures, 4 by 5



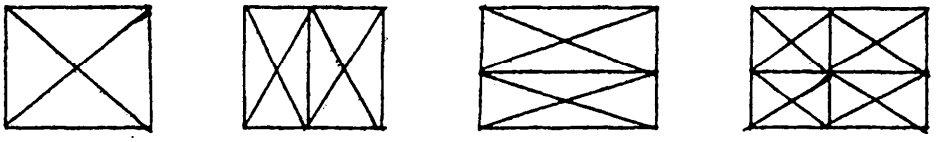


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Style (b):

9-1/2 by 9-1/2 inch - Nondirectional.

1 exposure, 9-1/2 by 9-1/2
2 exposures, 4-3/4 by 9-1/2, longitudinal with table.
2 exposures, 4-3/4 by 9-1/2, transverse to table.
4 exposures, 4-3/4 by 4-3/4



3.5.7.3.7.5 Overlap and spacing. Each radiograph shall be centered to the geometric center of the fluoroscopic screen within 3/16 inch. In any combination of more than one exposure on a single film listed in 3.5.7.3.7.4 overlapping of exposures or spacing between exposures shall not exceed 3/16 inch.

3.5.7.3.7.6 Semiautomatic movement. A semiautomatically operated cassette movement is one in which the cassette is moved from its standby position to the radiographic position by levers, linkages, springs, air-checks and similar devices when activated by the operator pushing a buttom or turning a lever and when the operator furnishes some of the energy for movement by returning cassette to standby position after exposure. It may or may not be necessary to shift or turn the cassette during the series to obtain certain of the combinations specified in 3.5.7.3.7.4.

3.5.7.3.7.7 Automatic movement. An automatically operated cassette movement is one in which the cassette is moved from its standby position to the radiographic position and returns by power drive, actuated by the operator pushing a buttom or turning a lever. It shall not be necessary to turn or shift the cassette during the series of exposures. This cassette movement shall also provide for the easy insertion and removal of the cassette from the device with a protective gloved hand, in the dark and without excessive reaching or inconvenience to either the patient or the operator. Type II, style (a) tunnels shall be arranged for either top or front loading of the cassette, and shall be provided with a manual ejector to assist in removal of the cassette. Type II, style (b) tunnels shall be arranged for front loading of the cassette, and shall be provided with an ejector which shall release the cassette and extend its leading edge to facilitate removal.

3.5.7.3.7.8 Controls. Each spotfilm tunnel shall be provided with all controls, exposure pattern selector, etc., to provide for successive movement of the cassette from the standby position to the spotfilm radiographic position. In addition, type II spotfilm tunnels shall be arranged to prevent a multiple exposure on the same area of the x-ray film when two or more "spots" are made on a single film.

3.5.7.3.7.9 Exposure. In tunnels providing for the automatic movement of the cassette it shall not be possible to make a spotfilm exposure until the cassette is in its properly centered spotfilm position. Also, it shall be possible to obtain the correct spotfilm radiographic exposure one second after the cassette movement is initiated without movement of the cassette tunnel or blurring the resultant radiograph. When the high speed rotating anode tube is used, within 2-1/2 seconds is acceptable before exposure is made.

3.5.7.3.7.10 Lead protection. The spotfilm tunnel shall be provided with sufficient equivalent lead protection at 150 KVP, 5 MA to protect the film in the cassette when the cassette is in its standby position during fluoroscopy. The hand of the operator controlling the shutters shall be protected by 1.75 mm. of equivalent lead protection at 150 KVP. A paneled flexible protective apron shall be supplied which shall be adjustable on at least 2 sides of the spotfilm device.

3.5.7.3.7.11 Phototimer receiver. When spotfilm phototiming is required by the invitation for bids, the spotfilm tunnel shall be equipped with a phototimer receiver which integrates illumination and is installed in such a manner that red-light or white-light illumination in the room will not effect the proper operation of the phototimer circuits. The phototimer receiver shall not interfere with the visual observation of the fluoroscopic screen or image intensifier during either fluoroscopy or radiography. Provision shall be made for the phototimer to be operable when an image intensifier is adapted to the spotfilm tunnel and is in position.

3.5.7.3.7.12 Phototimer density control. When the spotfilm tunnel is equipped for phototiming, a density control convenient to the operator's position while using the apparatus shall provide for varying the density of exposed radiographs, as described in 3.3.3.2.4.4.



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3.5.7.3.7.13 Spotfilm tunnel fluoroscopic screen. The fluoroscopic screen shall be mounted in the spotfilm tunnel to provide a useful area of at least 144 sq. inches, and having one dimension not less than 12 inches. It shall be mounted under lead impregnated glass providing at least 2.3 mm. of equivalent lead protection at 150 KVP. The fluoroscopic screen shall consist of fluorescent crystals in a suitable plastic binder either coated on a base of flat flexible plastic sheet or incorporated in a plastic medium to form a flat flexible sheet. The screen shall be uniformly radiolucent, containing no inclusions of metal, fiber, or other relatively radiopaque foreign substances which might cast a shadow on the fluorescent image. It shall be moisture-proof and of the cleanable type. The chemical shall fluoresce under x-radiation in the kilovoltage range normally employed in medical fluoroscopy. The spectral distribution of the fluorescence shall be within the maximum sensitivity range of the eye with the peak emission between 520 and 560 millimicrons. The screen shall have no appreciable after-glow which would cause loss of image sharpness when observing fluoroscopically a moving part such as the heart.

3.5.7.3.7.14 Accessory tracks. Tracks or other devices shall provide for the mounting of a radiographic grid or compression cone to the tunnel. These tracks or other devices shall not be visible upon the fluoroscopic screen. Latches shall be supplied to hold accessories in place.

3.5.7.3.7.15 Radiographic grid. The retractable radiographic grid shall be of the 80 line per inch and 8 to 1 ratio type of focused grid. It shall be arranged so that it can be moved into or out of the radiographic field at the discretion of the operator. The grid shall cover the entire area of the fluoroscopic screen. The focal length of the grid shall be proper to cover the target-to-film distances possible in spotfilm radiography.

3.5.7.3.7.16 Compression cone. A rectangular cone which shall limit the field to a 4 by 5 inch area shall be supplied with style (a), (8 by 10) spotfilm tunnels, which the operator must orient with the proper axis of the film. A square compression cone shall be furnished with style (b) spotfilm tunnels to limit the exposure areas as prescribed in 3.5.7.3.7.4 and 3.5.7.3.7.5. When attached to the accessory tracks, it shall be possible, at the operator's discretion, to move the cone into or out of the radiographic field. A latching device will hold the cone in either of the above positions. When the cone is in the radiographic position, the central ray of the x-ray tube shall travel through the center of the compression cone within 1/4 inch.

3.6 Body-sectional devices.. When specified in the invitation for bids, a remote-controlled, power-driven, body-sectional device of the type specified shall be furnished.

3.6.1 Plane selector. A means shall be provided for indicating and selecting the plane parallel to the tabletop of which a radiograph is to be made. This plane selector shall be calibrated and adjustable in 1/2 cm. increments and cover the range of 2 through 25 cm. The calibration marks shall indicate the distance of the plane from the top of the table and as calibrated in 4.4.3.15.

3.6.2 Focal spot and bucky travel. The focal spot of the x-ray tube shall travel through the chord of an arc during horizontal or vertical body-sectional radiographic procedure. During the x-ray tube movement, the bucky diaphragm shall travel in opposite directions. The movements of the x-ray tube and bucky diaphragm shall be synchronized in such a manner that the resultant radiograph indicates the image of the plane indicated on the plane selector.

3.6.3 Tabletop coverage. It shall be possible to operate the body-sectional device anywhere along the length of the tabletop where the bucky diaphragm can complete the movement required of it for any plane or planes selected for a given series of body sectional radiographs. When the body-sectional device cannot be operated anywhere along the length of the tabletop, but only in a fixed position, a tabletop extension shall be furnished for positioning of the patient. The fluoroscopic assembly shall not limit the movements when it is properly parked and when a power-driven tabletop is specified it shall be in its normal centered position.

3.6.4 Amplitude selector. The amplitude selector shall control the useful range of travel of the x-ray tube and shall be adjustable to provide the ranges of thickness of section in the increments specified. The amplitude selector shall include provisions for starting and stopping of the driving mechanism, and for controlling the x-ray exposure during the tube excursion. Provision shall be made to prevent blurring of radiographs due to vibrations. A suitable means for determining the thickness of section shall be provided to permit adjustment of the thickness in approximate 2 mm. increments within the range of 2 through 6 mm. Any charts shall be attached to the apparatus. The performance shall be determined as described in 4.4.3.14 and shall meet the stated requirements.

3.6.5 Tube excursion. The x-ray tube excursion speed shall be uniform at any given setting and shall provide for speeds through the exposure range from a minimum of at least 3-1/2 inches per second to a higher speed of at least 7-1/2 inches per second.

3.6.6 Target-to-film distance. The body-sectional device shall have a target-to-film distance within the range of 36 to 40 inches.



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3.6.7 Driving mechanism. For ceiling-mounted tube stands, no part of the body-sectional apparatus shall contact the floor.

3.6.8 Remote control. The control for initiating the operation of the body-sectional device shall be located in or in the vicinity of the x-ray control cabinet. The remote control shall provide for the coordinated operation of the x-ray tube, bucky diaphragm, body-sectional driving mechanism and x-ray exposure. The operation shall be initiated by a dead-man type of switch or through the radiographic timer initiating means (see 3.3.3.1.9) so that the operator may "stop the mechanism" at any time.

3.6.9 Removability. The body-sectional devices shall be removable from the table and tube stand or shall be of such design as not to be objectionable when the x-ray apparatus is used for other types of radiological examinations. The removable portions shall not require the use of tools. Stretchers and personnel must be able to move freely around the table.

3.7 X-ray tubes. Radiographic and fluoroscopic x-ray tubes shall be of the rotating anode shockproof type employing shockproof cable receptables as per Fed. Std. No. 72. They shall be equipped with blowers to provide the specified overall cooling rate. The temperatures of any external part of the housing shall not exceed 175° F. when tested as specified in 4.5.

3.7.1 X-ray tube ratings.

3.7.1.1 Class 1 x-ray tubes shall have a minimum rating of 125 KVP.

3.7.1.2 Class 2 x-ray tubes shall have a minimum rating of 150 KVP.

3.7.2 Stator and rotor.

3.7.2.1 Class 1 x-ray tube stator and rotor. The rotor shall attain a speed from rest in 1 second to permit safe exposures within the rating of the tube and with apparatus described for its use herein.

3.7.2.2 Class 2 x-ray tube stator and rotor.

3.7.2.2.1 Style (a) Conventional inserts. The requirements for class 2, style (a) tube stator and rotors shall be as stated in 3.7.2.1.

3.7.2.2.2 Style (b) high-speed inserts. The rotor of high-speed tubes shall attain a speed from rest of at least 8500 r.p.m. within 2-1/2 seconds when 220 volts are applied to the stator through suitable frequency changing equipment. It shall be possible for the rotor to maintain a speed of at least 8500 r.p.m. with 110 volts applied on the rotor (see 3.3.3.1.11 (c) and 4.4.1.13).

3.7.3 Low-voltage cords and clamps. Cords shall supply power for the stator, blower, and radiographic field illuminator or central ray indicator. Clamps shall be provided for clamping the low-voltage cables to either of the shockproof cables. These clamps shall have no sharp edges, and shall be so designed that they cannot damage the shockproof cables.

3.7.4 Ports. The x-ray tube unit port shall be transparent to permit visual inspection of filament and target. The center of the focal spot of the insert shall be within 1/16 inch of a line which is perpendicular to the plane of the tube port mounting plate, and which goes through the center of the port. The port and the insert shall be so designed that the x-ray beam coverage will be at least 17° from the central ray in all directions.

3.7.5 Inserts. The x-ray tube insert shall be adequate to permit performance as required herein without damage to the insert or its housing. The filaments shall be of pure tungsten wire of such design as not to require more than 5.5 amperes of current at 13 volts for the MA and KVP for the focal-spot sizes in the following tables:

TABLE IV. Filament characteristics

| Type I or II, class 1 x-ray tubes | | |
|---|---|---|
| Focal-spot size (nm.) | MA | KVP |
| 0.3 . . . . . | 40 | 50 |
| 0.7 . . . . . | 100 | 60 |
| 1.0 . . . . . | 200 | 60 |
| 1.5 . . . . . | 300 | 60 |
| 2.0 . . . . . | 500 | 50 |



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TABLE V. Filament characteristics

| Type I or II, class 2 x-ray tubes | | |
|---|---|---|
| Focal-spot size (mm.) | MA | KVP |
| 0.5 | 85 | 65 |
| 0.7 | 100 | 65 |
| 1.0 | 200 | 65 |
| 1.5 | 300 | 70 |
| 2.0 | 500 | 70 |

3.7.5.1 Focal-spot size. The area of the focal-spots shall be measured as prescribed in Fed. Std. No. 83, with the center of the small focal spot being located within the periphery of the large focal spot. Unless otherwise specified in the invitation for bid, the focal-spot sizes and combination of sizes shall be standard commercial production, as listed in the following table (see 6.1 (v)).

TABLE VI. Focal-spot sizes

| Focal-spot size (mm.) | |
|---|---|
| Class 1 tubes | Class 2 tubes |
| 0.3 to 1.5 | 0.5 to 1.5 |
| 0.7 to 2.0 | 0.7 to 2.0 |
| 1.0 to 2.0 | 1.0 to 2.0 |
| 1.5 to 1.5 | |

3.7.5.2 Target disk. A solid tungsten or combination rhenium-tungsten and molybdenum or molybdenum-tungsten disk anode shall be required. The size of the disk, speed of rotation and the diameter of the focal ring shall be such that the ratings shown in the following tables are within safe operating limits of the disk 7-1/2 minutes after being heated to full anode rating.

TABLE VII. Focal-spot ratings

| Class 1 x-ray tubes | | | |
|---|---|---|---|
| Focal spot (mm.) | KVP | Second | MA |
| 0.3 | 125 | 1/10 | 17 |
| 0.7 | 125 | 1/10 | 75 |
| 1.0 | 125 | 1/10 | 160 |
| 1.5 | 125 | 1/10 | 245 |
| 2.0 | 125 | 1/10 | 330 |
| 0.3 | 100 | 1/4 | 20 |
| 0.7 | 100 | 1.0 | 70 |
| 1.0 | 100 | 4.0 | 100 |
| 1.5 | 100 | 8.0 | 100 |
| 2.0 | 100 | 10.0 | 100 |

TABLE VIII. Focal-spot ratings

| Class 2, style (a) x-ray tubes | | | |
|---|---|---|---|
| Focal spot (mm.) | KVP | Second | MA |
| 0.5 | 150 | 1/30 | 45 |
| 0.5 | 150 | 1/2 | 30 |
| 0.7 | 150 | 1/5 | 60 |
| 0.7 | 150 | 4 | 40 |
| 1.0 | 150 | 1/20 | 150 |
| 1.0 | 150 | 1-1/4 | 100 |
| 1.5 | 150 | 1/10 | 200 |
| 1.5 | 150 | 3/4 | 150 |
| 2.0 | 150 | 1/15 | 300 |
| 2.0 | 150 | 1.0 | 200 |
| 0.5 | 125 | 1/10 | 50 |
| 0.7 | 125 | 1/60 | 80 |
| 1.0 | 125 | 1/20 | 175 |
| 1.5 | 125 | 1/60 | 300 |
| 2.0 | 125 | 1/30 | 375 |



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TABLE IX. Focal-spot ratings

| Class 2, style (b) x-ray tubes | | | |
|---|---|---|---|
| Focal-spot (mm.) | KVP | Second | MA |
| 0.5 | 150 | 1/10 | 70 |
| 0.5 | 150 | 1-1/2 | 45 |
| 0.7 | 150 | 1/5 | 100 |
| 0.7 | 150 | 4 | 60 |
| 1.0 | 150 | 1/10 | 225 |
| 1.0 | 150 | 1/2 | 175 |
| 1.5 | 150 | 1/10 | 350 |
| 1.5 | 150 | 3 | 150 |
| 2.0 | 150 | 1/10 | 470 |
| 2.0 | 150 | 1 | 280 |
| 0.5 | 125 | 1/30 | 90 |
| 0.7 | 125 | 1/20 | 130 |
| 1.0 | 125 | 1/20 | 300 |
| 1.5 | 125 | 1/4 | 350 |
| 2.0 | 125 | 3/40 | 575 |
| 2.0 | 125 | 3/10 | 475 |
| 2.0 | 125 | 1.0 | 325 |

3.7.5.3 Bearings. The bearings shall be quiet in operation. The life of the bearings shall be approximately equal to the filament life. There shall be no excessive vibration of the tube unit. When class II, style (b) high-speed inserts are specified, a brake shall be furnished to rapidly reduce the rotor speed after each exposure. The time required to reduce the speed from 9000 to 3000 r.p.m. shall not exceed 3 seconds.

3.7.6 Thermal ratings. The thermal rating of the anodes and x-ray tube units (insert plus housing) shall be not less than indicated in the table below:

TABLE X. Thermal ratings

| | For class I units | For Class II, style (a) units | For Class II, style (b) units |
|---|---|---|---|
| Anode heat storage capacity in heat units. | 110,000 | 135,000 | 170,000 |
| Anode cooling rate, maximum in heat units per minute. | 30,000 | 30,000 | 30,000 |
| Total heat storage capacity in heat units. | 1,250,000 | 1,250,000 | 1,250,000 |
| Maximum overall cooling rating in heat units per minute. | 30,000 | 30,000 | 30,000 |

**NOTE:** Tube units shall be tested for performance and thermal capacity as described in 4.5.1 and 4.5.2.

3.7.7 Identification marking. Each shockproof housing unit shall be permanently marked with a nameplate showing manufacturer's model and type, focal-spot sizes, serial number, and the name or registered trademark of the manufacturer. The tube unit shall be marked "Equipped with Federal Standard Terminals". The terminals shall be marked to indicate which is "ANODE" and "CATHODE". The markings shall be legible and permanent.

3.7.8 Kilovoltage rating. Class 1 x-ray tubes shall be rated for operation up to a maximum of 125 KVP full wave rectified. Class 2 x-ray tubes shall be rated for operation up to a maximum of 150 KVP full wave rectified. The tube units shall withstand a high voltage use of 105 percent of the rated KVP when tested as specified in 4.5.1 (b). Any evidence of loss of vacuum in the insert, or electrical breakdown in the insert or housing such as gas glow in the insert, fluctuation of MA during test which can be attributed to gas, cracking, or snapping sounds, etc., shall be considered evidence that tube unit is unsatisfactory.

3.7.9 Radiographic x-ray tube unit. The physical design of the radiographic x-ray tube unit shall be such that when mounted upon tube-stands, none of the tube-stand's movements are limited, restricted, or hampered.

3.7.10 Fluoroscopic tube unit. The physical design of the fluoroscopic tube unit shall be such that when mounted to the fluoroscopic assembly of the table, none of the prescribed movements or requirements of the assembly are limited, restricted, or hampered.



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3.8 Cables. All cables required for operation shall be supplied in whatever types, sizes, or lengths as necessary to assure proper functioning of the complete x-ray equipment assembly, and including allied apparatus when specified.

3.8.1 Shockproof cables (including their terminals). Shockproof high-tension cables shall be measured excluding their high-tension terminals. The total length, unless otherwise specified in the invitation for bid, shall be such as to provide for the proper operation of the apparatus when installed in a proper neat and workmanlike manner.

3.8.1.1 Rating. The shockproof cables acting together shall be capable of withstanding 125 percent of the rated voltage of the x-ray tube to which they are connected for 1 minute and charged by the full wave rectified generator with which they will be utilized.

3.8.1.2 Construction. The cables shall be so constructed that repeated bending in any installed position will not be unduly harmful to them.

3.8.1.3 Cable terminals. Each cable shall be terminated with plugs for insertion into shockproof cable receptacles of high-tension transformers or x-ray tube housing and be in accordance with Fed. Std. No. 72. The shockproof cables shall be so arranged as to permit their use as either a cathode or anode cable and to permit their terminating plugs to be used interchangeably in the high-tension transformer and x-ray tube housing. When specified in the invitation for bid, a special 4 pin cathode cable shall be supplied to permit the use of grid control equipment (see 6.1 (bb)).

3.8.2 Strain relief. Strain relief shall be provided to relieve the bending stress on the cables where necessary in their installed position. These strain relief sleeves shall be provided with proper fastening devices.

3.8.3 Cable supports. Cable supports and guides shall be provided where necessary to prevent the installed cables from interfering with the operation of the equipment and to prevent excessive bending or chafing of the shockproof cables. The cables, when installed, shall present a neat and workmanship job (see 6.1 (w)).

3.9 Interchangeability of parts. All individual parts or assemblies of parts of any unit supplied on a contract shall be interchangeable with the corresponding parts or assemblies of parts of all other like complete units supplied in the same contract.

3.10 Service data. The contractor shall furnish with each unit two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts. This information will specifically include instructions, drawings, and templates for plumbing, power wiring, structural modifications, and special support devices necessary for roughing-in preparations prior to installation of the unit. The service data shall include such detailed information which will permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information.

3.10.1 Wiring diagrams. Wiring diagrams shall be included for electrical items or components. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts may be ordered by the manufacturer's part number, and without confusion as to the nomenclature. Identical or interchangeable replacement parts shall be identified by a single part number only.

3.10.2 Service data envelopes. When specified in the invitation for bid, one set of the service data, as indicated above, will be placed in a sealed waterproof envelope, and affixed securely to the outside of the shipping container. The envelope shall be secured to the shipping container by use of metal ties, nails, or similar means, and shall be located at the end of the shipping container, preferably, when space permits, at the end bearing the markings required by MIL-STD-129. In cases where complete service data is too bulky to be conveniently attached to the outside of the shipping container, the contents of the exterior envelope may be limited to information necessary for roughing-in preparations and a note shall be made on the exterior envelope which will indicate where the remainder of the first set of service data will be located (see 6.1 (x)).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements (see 6.3).



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4.2 High potential requirement. The high potential requirement of 3.2.2.1 shall be ascertained as follows:

(a) Proceed as outlined in UL No. 187, paragraphs 144 through 150.
(b) This test may be waived when proof is submitted that the type of x-ray transformer to be supplied has been listed by the Underwriters' Laboratories, Inc.

4.3 Circuit breaker requirement. The circuit breaker requirement of 3.3.3.1.2 shall be ascertained as follows:

(a) Proceed as outlined in UL No. 187 paragraphs 159 through 163.
(b) This test may be waived when proof is submitted that the type of x-ray control cabinet to be supplied has been listed by the Underwriters' Laboratories, Inc.

4.3.1 High voltage rectifiers. The rectifiers may be removed from the high voltage circuit during tests of circuit breaker, but if this is done the control settings shall be altered to increase the primary current drawn under these conditions by an amount equivalent to the primary current required to operate the filaments of rectifier tubes when tubes are used.

4.4 X-ray generator test.

4.4.1 Test equipment required.

4.4.1.1 Recording oscillograph. Recording oscillograph shall be multiple channel, direct recording type that will record impulses directly on a moving chart. There shall be at least three chart speeds ranging between approximately 0.4 and 50 inches per second, so that impulses over the complete range of seconds and fractional seconds can be measured.

4.4.1.2 Cycle counter. Cycle counter shall be of the synchronous motor type that will accurately indicate cycles or fractional seconds. It shall have a sweep pointer indicating over a scale divided into impulses or to 0.01 seconds. It shall have a total time scale graduated to at least 30 seconds. It shall be operable from 115 volts, 60 cycle supply.

4.4.1.3 Oscilloscope. Oscilloscope shall be dual trace and capable of algebraic addition. Tektronix model 564 with memory feature or equivalent is preferred. Tektronix models 422 or 541A with P7 phosphor screens or equivalent are acceptable but difficult to use because a single sweep trace cannot be analyzed thoroughly before it fades out. In such case, a scope camera must be used.

4.4.1.4 **Voltmeter.** Voltmeter shall be iron vane type AC voltmeter of one of the following or equivalent, but meters which rectify the AC and apply it to the DC meter movement are not acceptable because readings are affected by the shape of the AC wave form.

a. Weston model 433 (Cat. No. 433-1906002) 150/300/600-V scales, 25 to 125 cycles, accuracy 3/4 percent of full scale.
b. Simpson model 10, 150/300/750-V scales, 25 to 125 cycles, accuracy 3/4 percent of full scale.
c. General Electric model AP-9, 150/300/600-V scales, 25 to 125 cycles, accuracy 3/4 percent of full scale.

4.4.1.5 Milliampere meter. Milliampere meter shall be fast acting, high-tension type DC milliampere meter of suitable accuracy, 2 percent of full scale or better.

4.4.1.6 Milliampere second meter. Milliampere second meter shall be high-tension type DC MAS meter of 2 percent accuracy of full scale or better, capable of reading at least 0 to 50 MAS and not more than 0 to 100 MAS.

4.4.1.7 Penetrometer. The penetrometer used in the radiographic comparison test shall consist of eleven steps of aluminum. The first step shall be 5 mm. thick and each of the 10 successive steps shall be 3 mm. thicker than the preceding step. The eleventh step shall be 35 mm. thick. The overall dimensions of the penetrometer shall be at least 8 inches long and 2 inches wide.

4.4.1.8 Plane calibration block. Plane calibration block shall be a wooden block or other suitable opacity material formed into the shape of an isosceles right triangle at least 2 inches wide. The hypotenuse surface shall be of one centimeter step design, with a lead numeral secured at each step, indicating the centimeter level from the tabletop from 1 through 25. Upon this hypotenuse surface shall be mounted a 24-gauge wire-mesh screen. Optional designs are acceptable that will offer the same results.

4.4.1.9 Phantom. Phantom shall be a presswood or masonite cube of approximately 5 inches, capable of adding or deleting thicknesses and approximating the average body density.



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4.4.1.10 Phantom for image intensifiers. Phantom for image intensifiers shall be of basic aluminum or satisfactory radiation absorbing material capable of demonstrating contrast, focus, line pair resolution, astigmatism, pin cushioning or barreling effect.

4.4.1.11 Linear scales. For linear scales, an eight foot metal tape is suitable.

4.4.1.12 High voltage bleeder. High voltage bleeder shall be two singles or one dual bleeder apparatus capable of being connected in the high voltage x-ray system, affording at least a 3 percent accuracy in selecting 0.001 of the voltage applied to the x-ray tube under test for graphic display. It shall not impair or alter the normal load characteristics of the x-ray system. An acceptable design follows:

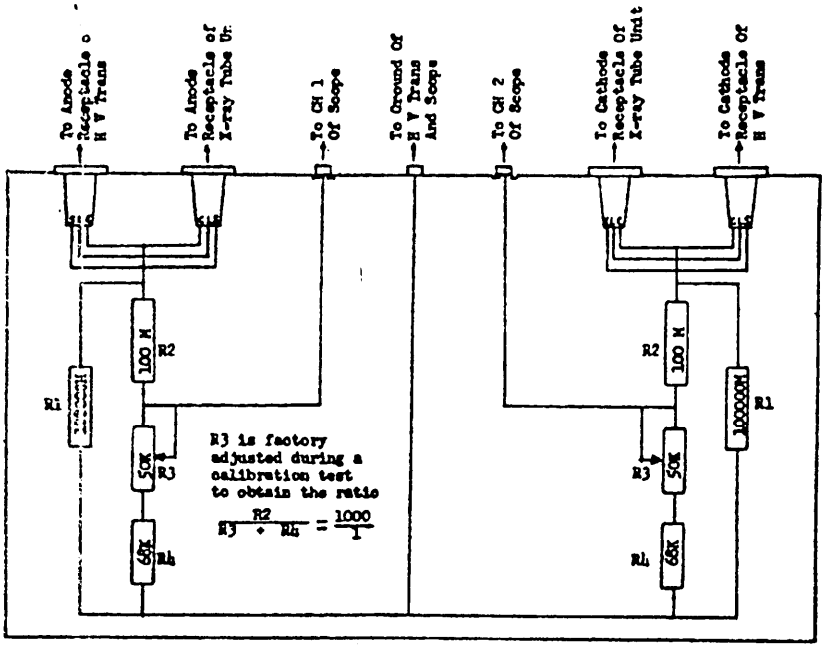



Fig. 1. HIGH VOLTAGE BLEEDER

4.4.1.13 Tachometer. A vibrating reed type tachometer or other suitable device shall be used to check compliance of 3.7.2.2.2 for tube rotational speed of 8500 r.p.m.

4.4.1.14 Sphere cap. Sphere cap shall be suitable for determining the distance at which ionization occurs between two spheres of a given size.

4.4.2 Turns ratio determination. The turns ratio of the high voltage transformer shall be originally determined from the manufacturer's drawings and as the test developes, later proven by the oscillograph or oscilloscope.

4.4.2.1 Calculated secondary KVP. From the turns ratio, calculate the "no load" primary volts required for 80 KVP using the formula:

$$\text{(Primary volts} = \frac{\text{KVP X 1000 X .707}}{\text{Turns Ratio}}\text{ .)}$$ Confirm with sphere-gap (see 4.4.1.14).

Disconnect the x-ray tube filament circuit and connect the voltmeter (see 4.4.1.4) across the primary to the high-tension transformer and make a 2-1/2 second simulated exposure to verify the turns ratio calculated accuracy. Record this voltage as it will be used to verify the calibration of the recording oscillograph to be connected later for simultaneous primary and secondary voltage recordings.

4.4.3 Performance tests. These tests shall be conducted with the aid of the high voltage bleeder system (see 4.4.1.12) in conjunction with the oscillograph (see 4.4.1.1) or the oscilloscope (see 4.4.1.3).

4.4.3.1 Bleeder and recorder connections. The H. V. bleeder consists of two voltage dividers calibrated so that the output voltage of each is 1/1000 of the input voltage. These are encased in a



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ten gallon drum filled with insulating oil for insulation between components and protecting the operator from the high voltages. The dividers are tapped off the H. V. cables of the x-ray unit by two connected H. V. receptacles through the bleeder cover. These are used as a tie point for the H. V. cables on the x-ray unit and 5-ft. jumper cables from the bleeder to the H. V. transformer or x-ray tube unit.

The schematic below illustrates the connections and use of the bleeder. Make connections as shown:

NOTE: While this arrangement is satisfactory for a scope of the specified models, consideration must be given to the use of recorders due to their usually lower input impedance to prevent overloading. Some modification to the input matching network may be necessary to include a pre-amplifier. This is acceptable as long as the calibration can result in the same degree of accuracy as that obtained with the scope.

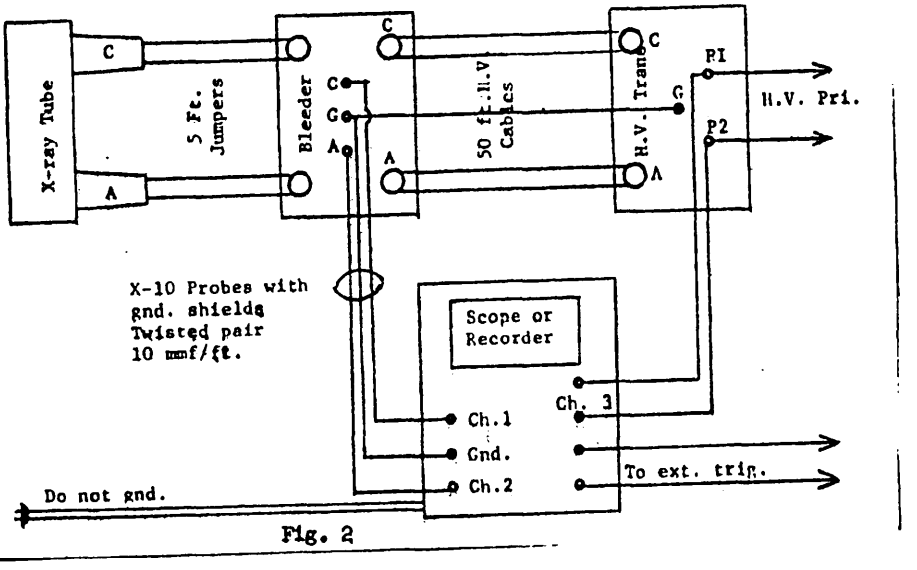



Fig. 2

4.4.3.1.1 Recorder settings.

H: 10-ms/div.
V: 2-v/div. x 10 for probe x 1000 for bleeder.

SELECT: ALG ADD
NO LOAD KVP AND TURNS RATIO TEST:

4.4.3.2 Make one or more simulated exposures and observe the tracing, disregarding the first peak. If this test indicates more than plus or minus 3 percent error, it may be due to the test apparatus, such as (a) H. V. jumper cables being more than 5 ft. long, (b) cable between scope probes and bleeder has too much capacitance (is other than a shielded, twisted pair), (c) scope is not correctly calibrated (d) bleeder is not correctly calibrated. This error may be due also to the x-ray unit H. V. cables being over 5 ft. longer or shorter than the prescribed 50 ft. length. After rechecking factors (a) to (d) inclusive, if the error is not over 5 percent, apply a percentage correction to each KVP reading made thereafter, with that particular test setup. If this error exceeds 5 percent, the test shall be terminated, and if the cause is determined to be the x-ray apparatus, this shall be cause for rejection.

4.4.3.3 Transformer electrical balance test. Set the x-ray timer for 1/20 second and the recorder for chopped mode operation. Make a simulated exposure and observe the tracing which shall show the total voltage to be within 3 percent of 80 KVP and each half to be within 2-1/2 percent of 1/2 of the total KVP. (If an unbalance exists, it can be demonstrated by reversing the channel leads. Paired cables must be of equal electrical characteristics.)

4.4.3.4 Contactor phasing and operation test. From the above tracing (see 4.4.3.3), an analysis will reveal the phasing of the contactor, i.e., "making" and "breaking" smoothly within 1.5 milliseconds of "zero". Distorted wave forms indicate improper performance. Since this is affected somewhat by the x-ray load, it will be necessary to again observe this later under load.

4.4.3.5 Timer test. Reset the recorder for "ALG-ADD", cover the x-ray tube port, reconnect the x-ray tube filament and conduct the timer tests using 20 percent of the available MA and 50 percent of the KVP and make exposures for the various timer settings and observe the pulse display to conform



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to the requirements of 3.3.3.1.9, 3.3.4.3, and 3.3.5.3.

4.4.3.6 MA meter test. Conduct the MA meter test for the requirements of 3.3.3.1.8 and 3.3.5.2 as follows:

Insert the milliampere meter (see 4.4.1.5) in the anode lead of the x-ray tube, in place of the anode bleeder, and check each scale at full scale, 0.7 of full scale and 0.4 of full scale, or at approximately equivalent MA technique selector positions.

4.4.3.7 MAS meter test. Replace the milliampere meter with the MAS meter (see 4.4.1.6) and check requirements of 3.3.3.2.2. The accuracy of the MAS meter shall be checked at 10, 25, and 50 MAS for exposures within a time range not exceeding 0.3 seconds.

4.4.3.8 Transformer regulation test. Remove the MAS meter from the anode circuit and reconnect the bleeder. Make exposures at each MA setting from 50 to maximum at 60, 80, 100, 120, and 150 KVP at 0.05 seconds within the ratings of the x-ray unit under test. From observing the voltmeter across the primary immediately prior to exposure, and the resultant tracings, chart the information in table XI.

NOTE: If the test is being conducted with the oscilloscope, it will be necessary to repeat the above exposures with the scope and X10 probe across the primary to arrive at the "load on" primary voltages. The "load on" primary volts should be corrected for the voltage drop within the silicon controlled rectifier or thyrotrons when an electronic x-ray contactors is used.

TABLE XI. Regulation data

Mfr._____ Model_____ Turns ratio_____

| MA | Primary volts | | KVP indicator | KVP recorder |
|---|---|---|---|---|
| | No load | Load on | | |
| | | | | |

4.4.3.8.1 Regulation graph. From the data contained in table XI, (see 4.4.3.8) plot graph with "load on" primary volts (abscissa) vs. KVP (ordinate) for the MA loads recorded.

The graph should indicate a slight deviation from parallel lines as the load increases. The regulation factor shall not exceed 6 KVP per 100 MA load at any of the check points indicated on the graph. The wave shapes shall be uniform and free of transient spikes.

4.4.3.9 Kilovoltage indicator test; including load compensation. Accuracy of the kilovoltage indicator can now be determined from table XI for conformance to 3.3.3.1.7.

4.4.3.10 Space charge compensation test.

4.4.3.10.1 For types I and II high-tension transformers.

(a) With a control cabinet setting of 100 MA, 50 KVP, make a 1/5 second exposure and note the MAS meter reading.
(b) Change the KVP to 100 with other factors remaining constant and repeat the exposure.
(c) Repeat the procedure for paragraphs (a) and (b) for each MA setting.
(d) A tolerance of no more than 5 percent differential is acceptable.

4.4.3.10.2 For types III, IV, V, VI, and VII transformers:

(a) For each MA setting as above (see 4.4.3.10.1) make two exposures, one at 60 KVP and the other at 125 KVP (above 500 MA, use 110 KVP for upper limits) while maintaining a timer setting to produce MAS in the range of 25 to 35. A differential within 5 percent is acceptable.

4.4.3.11 Timer recycling. The requirements of 3.3.5.3.1 shall be checked with the recording oscillograph connected across the primary leads of the high-tension transformer when an angiographic apparatus capable of repetition rate prescribed is connected to the apparatus.

4.4.3.12 Radiographic comparison test. The functioning of the high-tension transformer and allied equipment shall be such that the deviation of a series of penetrometer exposures for each technique selector position for a given kilovoltage shall not exceed one penetrometer density step. Penetrometer (see 4.4.1.7) exposures shall be made at 60, 100, and 150 KVP using medical screen type x-ray film with parspeed intensifying screens. (See 4.4.3.12.1 for suggested penetrometer techniques. If necessary, the equivalent values for other MAS combinations shall be obtained utilizing the inverse square law for distance.)



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4.4.3.12.1 Penetrometer techniques. The following penetrometer techniques are recommended for the comparison test. While the specific techniques are not mandatory, deviations resulting in equivalent results may be substituted.

TABLE XII. Penetrometer technique

| KVP | MA | Time in seconds | MAS | Distance |
|---|---|---|---|---|
| 60 | 50 | 1 | 50 | 72 inches |
| 60 | 75 | 3/5 | 45 | 68 inches |
| 60 | 100 | 1/2 | 50 | 72 inches |
| 60 | 150 | 3/10 | 45 | 68 inches |
| 60 | 200 | 1/4 | 50 | 72 inches |
| 60 | 300 | 1-1/20 and 1-1/10 | 45 | 68 inches |
| 60 | 500 | 1/10 | 50 | 72 inches |
| 60 | 600 | 1/12 | 50 | 72 inches |
| 60 | 800 | 1/15 | 53.3 | 74 inches |
| 100 | 50 | 1/5 | 10 | 93 inches |
| 100 | 75 | 2-1/15 | 10 | 93 inches |
| 100 | 100 | 1/10 | 10 | 93 inches |
| 100 | 150 | 1/15 | 10 | 93 inches |
| 100 | 200 | 1/20 | 10 | 93 inches |
| 100 | 300 | 1/30 | 10 | 93 inches |
| 100 | 500 | 1/60 | 8.33 | 84 inches |
| 100 | 600 | 1/60 | 10 | 93 inches |
| 100 | 800 | 1/120 | 6.6 | 75 inches |
| 150 | 50 | 1/30 | 1.66 | 100 inches |
| 150 | 75 | 1/30 | 2.5 | 124 inches |
| 150 | 100 | 1/60 | 1.66 | 100 inches |
| 150 | 150 | 1/60 | 2.5 | 124 inches |
| 150 | 200 | 1/120 | 1.66 | 100 inches |
| 150 | 300 | 1/120 | 2.5 | 124 inches |

4.4.3.13 Bucky grid radiographs. A series of strip radiographs shall be made on a single film in a cassette in the bucky tray using the technique tabulated below. Exposures shall be made through a penetrometer, with adjacent film areas protected by lead.

TABLE XIII. Bucky grid technique for super-speed bucky

| Exposure | KVP | MA | Time | Target film distance |
|---|---|---|---|---|
| 1 | 68 | 300 | 1/30 | 40 inches |
| 2 | 52 | 200 | 1/5 | 40 inches |
| 3 | 52 | 200 | 3/10 | 40 inches |
| 4 | 50 | 200 | 1/2 | 40 inches |
| 5 | 50 | 200 | 1 | 40 inches |

NOTE: (a) Use medical screen type x-ray film and medium speed intensification screens.
(b) For parspeed bucky tests, substitute 60 KVP, 200 MA, 1/10 second 40 inches target film distance in place of exposure No. 1.

4.4.3.14 Amplitude of section. The requirements of 3.6.4 shall be determined as follows:

(a) The plane calibration block (see 4.4.1.8) shall be used as the object for a body-sectional radiographic exposure. The film shall be examined and the number of visible wires counted. The count is an indication of the thickness of the slice depicted by the body-sectional radiograph.

(b) A 24-gauge wire mesh has squares whose dimensions are 24/25 mm. The number of wires sharply defined on radiographs taken as above shall be in accordance with the following table.

TABLE XIV. Thickness requirements

| Thickness of slice mm. | Number of wires sharply visible |
|---|---|
| 2.0 | 2 to 3 |
| 3.0 | 3 |
| 4.0 | 3 to 4 |
| 5.0 | 4 |
| 6.0 | 5 to 6 |



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4.4.3.15 Plane calibration. The requirements of 3.6.1 shall be determined as follows:

a. Using a thickness of section setting of 5.0 mm. set the plane selector for any level and make an exposure of the plane calibration block (see 4.4.1.8). The processed radiograph shall distinctly reveal the lead number image corresponding to the setting of the plane selector.

4.4.3.16 MA stabilization test. All units shall be calibrated and adjusted prior to each of the following tests in accordance with the manufacturer's instructions.

4.4.3.16.1 For types I and II high-tension transformers:

(a) With a control cabinet setting of 100 MA and approximately 60 KVP, note the milliammeter reading for a 1 second exposure.
(b) Adjust the line compensator device (see 3.3.3.1.4) for a 13 plus or minus 1 volt higher line voltage setting (6 percent) readjust the kilovoltage controls to 60 KVP and repeat the exposure.
(c) Adjust the line compensator device for a 13 plus or minus 1 volt lower line voltage setting, readjust the kilovoltage control to 60 KVP, and repeat the exposure.
(d) The readings of the milliamperage shall remain constant within 5 percent of that of the original setting per paragraph (a).

4.4.3.16.2 For types III, IV, V, VI, and VII high-tension transformers:

(a) With a control cabinet setting of 200 MA and approximately 60 KVP, note the MAS meter reading for a 1/10 second exposure.
(b) Adjust the line compensator device (see 3.3.3.1.4) for a 13 plus or minus 1 volt higher line voltage setting (6 percent) readjust the kilovoltage controls to 60 KVP and repeat the exposure.
(c) Adjust the line compensator device for a 13 plus or minus 1 volt lower line voltage setting, readjust the kilovoltage control to 60 KVP, and repeat the exposure.
(d) The readings of the MAS meter shall remain constant within 5 percent of that of the original setting per paragraph (a).
(e) With a control cabinet setting of 100 MA and approximately 60 KVP, note the milliammeter reading for a 1 second exposure.
(f) Adjust the line compensator device (see 3.3.3.1.4) for a 13 plus or minus 1 volt high-line voltage setting (6 percent) readjust the kilovoltage controls to 60 KVP and repeat the exposure.
(g) Adjust the line compensator for a 13 plus or minus 1 volt lower line voltage setting, readjust the kilovoltage controls to 60 KVP, and repeat the exposure.
(h) The readings of the milliammeter shall remain constant within 2 percent of that of the original setting per paragraph (e).

NOTE: If x-ray equipment being tested does not utilize a KV meter, it will be necessary to perform these tests using primary voltage readings.

4.4.3.17 Phototimer tests. When a phototimer is specified (see 3.3.3.2.4), it shall be tested as follows and preliminary to each of the following tests, the apparatus shall be calibrated as described in the manufacturer's literature to obtain proper radiographic results.

(a) Phototimer dark current test:

Disconnect the x-ray tube filament circuit and the primary of the high-tension transformer, also any safety exposure terminating or compensating circuits so that the phototubes will directly terminate the exposure through the phototimer. Make an exposure at each of the phototube receiver positions timing the interval by a cycle counter (see 4.4.1.2). The exposure as terminated by the dark current of the phototube shall be at least 10 times the normal exposure time required for the normal radiographic technique at each phototube receiver.

(b) Reproducibility of results. This feature shall be tested by making a series of 10 consecutive exposures through a phantom (see 4.4.1.9) for each of the phototube receivers. This phantom shall consist of 5-3/4 inches or more, if required, thickness of masonite or pressed wood. The technique factors chosen shall be such that the exposure time is within the range of from 0.2 to 0.3 seconds at a minimum of 60 KVP.

(c) Minimum response time. With settings of 200 MA at 60 KVP, make a series of exposures through phantom described above for each of phototube receivers. Adjust the phototimer density controls or remove sections of the phantom until a reading of 5.0 MAS (3 impulses) is obtained. Then increase the kilovoltage settings in steps until a reading of 3.3 MAS (2 impulses or 1/60 second) is obtained. Continue to increase the kilovoltage to a maximum of 125 KVP. The minimum response time shall not exceed 2 impulses or 1/60 second. This test shall be required only when the phototimer is installed.



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in a control provided with an impulse timer to function with types III, IV, V, VI, and VII high-tension transformers.

(d) Phototimer safety device. An exposure shall be made at the positions of the phototimer selector, i.e., with the phototimer in the circuit and with the x-ray tube or high-tension transformer disconnected. The exposure shall be terminated by the phototimer safety device prior to exceeding the safe time limit for the particular MA setting, focal spot size and maximum KVP to be used for phototiming; the safe time limit to be determined from the manufacturer's tube rating charts and exposure time measured by a cycle counter, or recording oscillograph.

4.5 X-ray tube tests. Tests may be made to determine compliance with the requirements of the specifications. At least the tests specified in 4.5.1 and 4.5.2 shall be run on a sample tube unit. To avoid undue wear, they need not be repeated for each x-ray tube unit offered for delivery by the manufacturer.

4.5.1 Performance test. Each tube unit shall be tested starting with a cold tube unit with not more than 10 second intervals between each step with the stator energized as described in 3.3.3.1.11, utilizing the large focal spot. For class 1 tubes:

(a) Apply 110 KVP, 50 MA for 20 seconds.
(b) Apply 125 KVP at 3 MA. Increase KVP smoothly (no sharp steps) to 105 percent of the manufacturer's rating at the rate of approximately 3 KVP per minute, maintaining 3 MA during the KVP increase. Then operate unit for 10 minutes at 3 MA at 105 percent of manufacturer's KVP rating.
(c) Operate tube unit for 20 minutes at 4 MA 125 KVP.
(d) Allow tube unit to cool for 7-1/2 minutes with fan off and stator voltage off.
(e) Apply 110 KVP 50 MA for 20 seconds.

For class 2 x-ray tubes:

(a) Apply 130 KVP 50 MA for 20 seconds.
(b) Apply 150 KVP at 3 MA. Increase KVP smoothly (no sharp steps) to 105 percent of the manufacturer's rating at the rate of approximately 3 KVP per minute, maintaining 3 MA during KVP increase. Then operate unit for 10 minutes at 3 MA at 105 percent of manufacturer's KVP rating.
(c) Operate tube unit for 20 minutes at 3 MA, 150 KVP.
(d) Allow tube unit to cool for 7-1/2 minutes with fan off and stator voltage off.
(e) Apply 135 KVP 50 MA for 20 seconds.

4.5.2 Heating test. The tube unit shall be operated in the horizontal position at 100 KVP and 5 MA continuously until equilibrium in temperature is reached. The air circulator shall be operating on a 115 volt, 60 cycle source, the rotor of the tube shall be stationary. Test results shall be corrected to an ambient condition at 30° C. (86° F.).

5. PREPARATION FOR DELIVERY

5.1 General. The requirements specified herein apply only to purchases by or direct shipment to the Government.

5.2 Packaging.

5.2.1 Level A.

5.2.1.1 Unit of issue. One x-ray apparatus, as specified, constitutes one unit.

5.2.1.1.1 X-ray tubeheads with tube. All openings in the tubeheads shall be sealed in such a manner as to prevent entry of any foreign substance. Each tubehead with tube shall be packaged in a corrugated fiberboard carton of suitable size and design. The carton design shall include suitable interior fittings or sufficient cushioning material to insure against shifting or breakage. Each carton shall be secured to prevent accidental opening. Cartons shall be overwrapped with waterproof barrier-material conforming to PPP-B-1055 and shall be sealed with an appropriate waterproof adhesive.

5.2.1.1.2 Rectifier tubes. Each rectifier tube shall be packaged in a carton of appropriate size and design. The carton shall be designed with interior fittings or shall contain sufficient cushioning material (cellulose wadding, excelsior, or other suitable material) to insure against shifting or breakage. The four cartons shall be overwrapped with waterproof barrier-material conforming to PPP-B-1055 and shall be sealed with an appropriate waterproof adhesive.

5.2.1.1.3 Control stand, table, transformer, and tube-stand. The control stand table, transformer, and tube-stand shall be disassembled as far as is practical. All exterior painted and finished surfaces shall be protected by wrapping with kraft paper and cushioning with excelsior pads, flexible



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corrugated fiberboard, or other suitable material. Wrapping and cushioning materials shall be secured by tying with cotton cord or tape, or sealing with kraft gummed-paper tape. The work shall be accomplished in such a manner as to insure against marring or scratching or damage to the unit. Bolts, nuts, screws, and miscellaneous small parts removed during disassembly shall be adequately protected and packaged by like kind. Parts shall be secured to or packed with the major item from which the parts have been removed.

5.2.1.1.4 Cables. Each cable shall be coiled of such diameter that the cable will not take a permanent set or that the outer covering will not be damaged. Each coiled cable shall be further protected with a spiral-wound wrapping or shall be packaged in a bag; wrapping or bag shall be fabricated from waterproof material conforming to PPP-B-1055.

5.2.2 Level C. Unless otherwise specified, the subject commodity shall be packaged in accordance with accepted commercial practice.

5.3 Packing.

5.3.1 Level A.

5.3.1.1 X-ray tubehead with tube and rectifier tubes. The tubeheads with tubes, and the package containing the four rectifier tubes shall be individually overpacked in an exterior container designed for type 3 load, and constructed in accordance with PPP-B-621, except that the minimum thickness of sides, tops, and bottoms shall be not less than 5/8 inch. Closure and strapping of boxes shall comply with the appendix to the box specification.

5.3.1.1.1 Cushioning. Boxes shall contain sufficient cushioning material to "float" the interior packages in such a manner that there shall be not less than 3 inches between all exterior faces of the interior packages and the inner faces of the exterior containers.

5.3.1.1.2 Waterproofing. Each box shall be lined with waterproof barrier material conforming to PPP-B-1055 and sealed with an appropriate waterproof adhesive.

5.3.1.2 Boxes for components weighing less than 1,000 pounds. Each of the above named components weighing less than 1,000 pounds shall be overpacked in an exterior container designed for type 3 load, and constructed in accordance with PPP-B-621. Closure and strapping of boxes shall comply with the appendix thereto.

5.3.1.2.1 Skids. If boxes, as specified above, are used for control unit, tube-stand, transformer, or table, then in addition, each box shall be fitted with skid of appropriate size to facilitate handling and to maintain the component in its proper position.

5.3.1.2.2 Bracing and cushioning. Components shall be secured within the shipping container by means of braces, blocks, or cushioning material in such a manner that the weight of the component will not be carried directly by the side and end of the exterior container, and to insure against shifting, breakage, or marring of surfaces.

5.3.1.2.3 Waterproofing. Each box shall be lined with waterproof barrier material conforming to PPP-B-1055 and sealed with an appropriate waterproof adhesive.

5.3.1.3 Crates for components weighing over 1,000 pounds. Part or parts weighing over 1,000 pounds shall be cushioned secured, and waterproofed as outlined above, except that each exterior container shall be a fully sheathed nailed wood crate constructed in accordance with MIL-C-104.

5.3.2 Level C. Unless otherwise specified, x-ray equipment shall be packed in containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Military agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.4.2 Civil agencies. In addition to any marking specified in the contract or order, packages and shipping containers shall be marked in accordance with the requirements of Fed. Std. No. 123.



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## 6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of high-tension transformer desired (see 1.2.1.1).
(c) Type of control cabinet. If the choice of one of the three types of control cabinets described herein does not meet the requirements of the user, detailed data relative to the user's requirements shall be given in the invitation for bids (see 3.3.2). The choice of one of the three types described will meet practically all users' requirements (see 1.2.1.2).
(d) Type of tube-stand (see 1.2.1.3).
(e) Type, class, and style of table (see 1.2.1.4).
(f) Type of body-sectional devices desired (see 1.2.1.5).
(g) Type, class, and style of x-ray tubes desired. Both type I and II tubes are required for type II, classes 3, 4, 5, and type III tables with tube-stands. A type I tube is required for types I and II, class 1 tables with tube-stands. A type II tube is required for type II, class 2 tables (see 1.2.1.6). Usually, class 1 tubes are specified for generators with types I, II, and III high-tension transformers, class 1 or 2, style (a) tubes for type IV high-tension transformers, and class 2, style (b) tubes for classes V, VI, and VII high-tension transformers.
(h) Whether more than one bucky will be used so the necessary selector switch will be supplied (see 3.3.3.1.14).
(i) Additional control cabinet components (see 3.3.3.2). It is necessary to evaluate what options will best meet your requirements and so state in the invitation for bids.
(j) Desired phototiming station(s) (see 3.3.3.2.4). If just the spotfilm receiver is included, then the phototimer will include circuitry for single station only.
(k) MA techniques (types I and II transformers) (see 3.3.4.1). The techniques provided should meet the user's requirements, but if not, the changes required should be carefully explained in the invitation for bids.
(l) Other timer requirements for types I and II apparatus (see 3.3.4.3.1). If requirements as specified do not meet user's needs, the required changes must be clearly specified in the invitation for bids.
(m) MA techniques (types III, IV, V, VI, and VII transformers) (see 3.3.5.1). The techniques provided should meet the user's requirements, but if not, the change required should be carefully explained in the invitation for bids.
(n) Other timer requirements for types III, IV, V, VI, and VII apparatus (see 3.3.5.3.2), if those described herein do not meet user's requirements. Body section work with low MA exposures may require a long exposure time when full excursions are employed for some anatomical parts for very thin sections. In some cases a faster recycling rate than 6 exposures per second may be required.
(o) Ceiling height of room in which type I tube-stand is to be installed (see 3.4.2.2).
(p) Additional radiographic area (see 3.4.2.3). If area covered by the tube-stand is not adequate for user's needs, specify his requirements in invitation for bids.
(q) Height of room in which type II tube-stand is to be installed (see 3.4.3.1).
(r) Accessories to be supplied (see 3.5.3.1).
(s) Parspeed bucky if it is to be substituted in place of the super speed type (see 3.5.5).
(t) Tilting range for type III tables - usually 90° (see 3.5.7.2.4).
(u) Class of spotfilm tunnel (see 3.5.7.3.7.3).
(v) Size of focal spots for both radiographic and fluoroscopic tubes. Be sure that they are capable of operation at the MA techniques furnished (see 3.7.5.1).
(w) Shockproof cable supports. A floor plan of the area in which the equipment is to be installed to help the bidder in determining the number of supports and length of cables required (see 3.8.3).
(x) Overseas shipment or warehouse storage, service data and where located (see 3.10.2).
(y) Levels of packaging and packing required (see section 5).
(z) Whether valve tubes or high-voltage semiconductors are to be supplied for type III, IV, V, or VI transformers. Type VII transformers will be supplied with semiconductors (see 3.2.2.2).
(aa) Whether the lower range of radiographic kilovoltage control should be included (types III, IV, V, VI, and VII transformers ) (see 3.3.3.1.6).
(bb) Special four-pin cathode shockproof cables, when grid control type x-ray tubes and equipment will be used (see 3.8.1.3).
(cc) Whether power assist is required on table (see 3.5.4).

NOTE: Contracting officers are advised to require in their invitation for bids that major components of an x-ray equipment bid such as transformers, controls, tables, and tube-stands shall all be awarded for aggregate grouping of these components all principally manufactured by the same manufacturer.

6.1.1 This specification is applicable for the procurement of radiographic, and fluoroscopic apparatus of ratings, types, and classes as covered in this specification and covers the complete unit or components.



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6.1.1.1 When purchasing a complete radiographic and fluoroscopic unit, the award must be made in aggregate covering all components required.

6.1.1.2 When a high-tension transformer is specified in the invitation for bids, a control cabinet must also be specified.

6.1.1.3 When a table is required, specify the type of table wanted, the degree of Trendelenburg tilt, the type spotfilm tunnel desired, and whether or not phototiming is required for general radiography and spotfilm work.

6.1.1.4 In specifying a floor-to-overhead tube-stand or an overhead mounted tube-stand, indicate the ceiling height and constructional details of the ceiling of the room in which this is to be installed.

6.1.1.5 Should only a single component be desired, other than a complete unit, the components with which the new required component is to function should be definitely indicated and it should also be stipulated that the component supplied shall be adapted to function properly with the existing apparatus.

6.2 X-ray concerns are not normally responsible for the installation of electrical service, ceiling and other alterations to support tube-stands and other equipment and radiation protection structural shielding. All power supply and wiring within the floors, walls and ceilings, and structural shielding together with other alterations are the responsibility of the purchaser. In this respect, the procuring activity does not know in advance who is to be the successful bidder, and in cases where new wiring, structural shielding or other alteration must be supplied by the user, the complete floor plans covering the installation should be part of the invitation for bid with the provision that the bidder will indicate in these plans the wiring or other alteration required for the functioning of the unit he proposes to furnish. Structural shielding shall be provided by the user.

6.3 The contracting officer may require inspection at the supplier's plant and the supplier shall provide all the material, equipment, and labor to make the tests and inspection necessary for the inspector to submit a report of his findings to the contracting officer. If the contracting officer decides that the equipment is to be inspected at a Government laboratory, the invitation for bids shall state the location of the laboratory. All shipping costs, assembly and disassembly of equipment shall be made by and at the expense of the equipment supplier. The testing laboratory shall not be responsible for any damage to the equipment being tested.

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